Spectral World Renderer: Unified 3D Hyperspectral Unmixing

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https://ming053l.github.io/Spectral-World-Renderer/

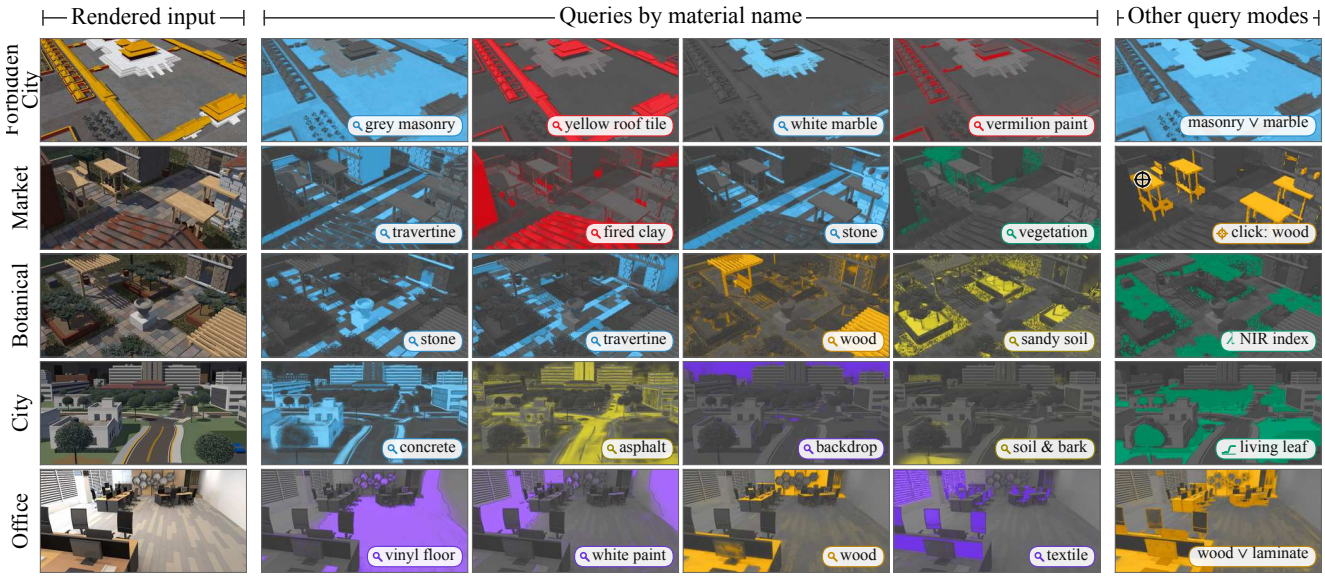


Figure 1. **Material queries in SWR-Bench worlds.** *Input:* multi-view hyperspectral images of worlds with known material composition, produced by the SWR world generator (left). *Output:* a Gaussian scene whose primitives carry material abundances and spectra. *Query:* a user selects a material by name (via a training-view map), a click (crosshair), a reference spectrum, a band index or a combination, and SWR colours the retrieved surfaces in any view; each tile shows one query on a test view.

Abstract

Cameras and the foundation models built on them recognize what a 3D scene contains, but not what it is made of: surfaces of the same colour can be different materials. Hyperspectral unmixing explains each spectrum as a mixture of material signatures, yet in 3D it is applied to rendered images after the views are composited, or tied to slow implicit fields, and no multi-view data with known composition exist to check the result. We present the Spectral World Renderer (SWR), which unifies world generation and unmixing. A coding agent builds executable 3D worlds that are rendered with measured spectra, so every hyperspectral image comes with exact labels and sub-pixel proportions; the resulting SWR-Bench provides eight indoor and outdoor worlds in which decompositions can be verified. SWR then unmixes inside rendering: each Gaussian of a compact 3D representation is decoded into material abundances before compositing, so one render yields spectra and material maps, and any surface can be queried by material name, a clicked pixel or a reference spectrum. Trained without material labels, SWR identifies materials best in all eight worlds, has the best mean PSNR on real captures, trains 3.7–12.9× *faster than unmixing baselines, renders 34.9 labelled views/s and separates same-coloured materials.*

1. Introduction

Knowing what a scene is made of, not only what it contains, matters for robotic sorting, agriculture and building inspection. Foundation models such as CLIP, SAM and DINOv2 recognize and segment objects from RGB images, and their features lift into 3D for open-vocabulary queries [38, 41, 60, 67]. They remain weak at identifying materials: three colours cannot separate surfaces of similar colour but different composition, such as green paint and a leaf; their labels name objects rather than materials, and one object may combine wood, laminate and metal; and they assign one category per pixel, like RGB material segmenters [4, 49, 84], whereas materials mix within a pixel.

Hyperspectral imaging supplies the missing evidence: the shape of a reflectance spectrum over tens to hundreds of bands characterizes the material [20, 22, 42, 53], and spectral unmixing explains each spectrum as a non-negative mixture of material signatures, or endmembers [5]. Extended to 3D, unmixing would reveal which materials each surface point contains, consistently across views.

This extension is difficult. (1) Hyperspectral data have many bands, so representations that store a spectrum per primitive grow with the band count. (2) Mixing occurs twice: within a pixel, as in 2D, and along each ray; after compositing, a mixture can no longer be attributed to the surfaces along the ray, and shading is entangled with visibility. (3) The decomposition is ambiguous: illumination changes the brightness and shape of a spectrum, and a model can explain two similar materials with one endmember while intensity absorbs their difference.

Existing methods treat reconstruction and unmixing as separate steps. Spectral radiance fields and Gaussian splatting [9, 24, 47, 55, 58, 76, 82] reconstruct spectra from per-band quantities or compact latents and are evaluated mainly by spectral fidelity; materials are then estimated by unmixing their renderings. UnMix-NeRF [63] unmixes inside a radiance field, but its implicit field makes training and rendering slow and lacks queryable primitives. Moreover, as Table 1 shows, the data needed to verify a decomposition are missing: real multi-view hyperspectral datasets are tabletop or turntable captures [9, 40, 44] that record radiance but no material labels or sub-pixel proportions, which cannot be annotated reliably by hand, and synthetic RGB datasets contain geometry and semantics, not spectra.

We therefore generate worlds whose composition is known by construction. In the world generator of our Spectral World Renderer (SWR), shown in Figure 2, a coding agent turns reference images and text into executable scene programs that fix geometry, materials, illumination and cameras. Measured spectra ground the materials, and shared sub-pixel samples render aligned RGB and hyperspectral observations, material labels and proportions. SWR-Bench offers eight worlds that separate fidelity from identity.

Spectral World Renderer adds spectra and materials to an RGB Gaussian representation and targets each difficulty. (1) An RGB-trained anchor provides compact geometry, and each Gaussian stores a band-independent feature instead of a spectrum, so storage does not grow with the bands. (2) A shared decoder maps this feature, at every ray intersection, to abundances over a scene-wide endmember dictionary, so mixing is resolved before compositing and one render yields both. (3) A separate intensity absorbs shading, and a multi-view appearance regularization keeps it from merging materials by assigning over-complete endmembers to distinct appearances; training uses no material labels. Per-Gaussian abundances answer queries in 3 ms by click, reference spectrum, band or a name attached through a training-view map. SWR has the best mIoU in every world and trains 3.7–12.9× faster than unmixing baselines on City. Known composition also exposes what fidelity hides: in Market and Study, 3DGS with post-hoc unmixing reaches a lower spectral angle yet far lower mIoU. Our contributions are:

Table 1. **Multi-view datasets with ground truth.** ID: material labels; Prop.: geometric area proportions; Gen.: worlds on demand. NeSpoF-Seg.: synthetic segmentation truth from UnMix-NeRF [63]. Only SWR-Bench samples proportions with each image and models spectral variability. –: not fixed or not stated. *We use 5 aligned scenes (Table S7).

Dataset	Bands	nm	Scenes ×views	Capture	ID	Prop.	Gen.
Replica	3	RGB	18×–	indoor	✗	✗	✗
Hypersim	3	RGB	461×168	indoor	✗	✗	✗
Kubric	3	RGB	–	random	✗	✗	✓
NeSpoF	21	450–650	4×18–30	tabletop	✗	✗	✗
NeSpoF-Seg.	21	450–650	4×–	synthetic	✓	✗	✗
HS-NeRF	128/141	371–1087	8×48–433*	turntable	✗	✗	✗
HSI-SC	204	397–1004	3×60	turntable	✗	✗	✗
SWR-Bench	64	420–980	8×96–240	in/outdoor	✓	✓	✓

- The SWR world generator and SWR-Bench: an agent writes executable worlds whose shared samples render spectra, labels and proportions, giving eight worlds with measured spectra that make decompositions verifiable.
- Spectral World Renderer, a compact Gaussian representation with band-independent storage that decodes mixtures at every intersection, so one render yields spectra and queryable 3D abundances.
- A multi-view appearance regularization that clusters training pixels across views and binds each cluster to its own endmember, without material labels.
- Real-time material queries by name, click, reference spectrum or band, consistent across views, that separate RGB metamers and resist shadow.

2. Related Work

Hyperspectral 3D reconstruction. Radiance fields use neural or grid models [2, 56, 57]; 3DGS composites Gaussians [37], later improved in quality and speed [25, 32, 52, 71, 91]. Foundation-model features [41, 67] answer object queries [8, 38, 66, 89, 95] and select materials in RGB [17], and neural harmonic textures vary appearance within a primitive [12]. Hyperspectral fields render radiance per wavelength [9, 40, 44, 47]; Gaussian versions store per-band quantities or compact latents decoded to spectra [24, 55, 58, 76, 82]. RGB inverse rendering recovers reflectance and illumination [6, 50, 93] or reflection components [85]. These methods model appearance, spectra or objects; SWR decodes each intersection into abundances and intensity, from features varying within each primitive.

Spectral unmixing. Unmixing explains each spectrum by endmembers and non-negative abundances [5, 26, 39], found by geometric extraction [59, 87], factorization or autoencoders [46, 62, 79], with spatial regularization [35] or

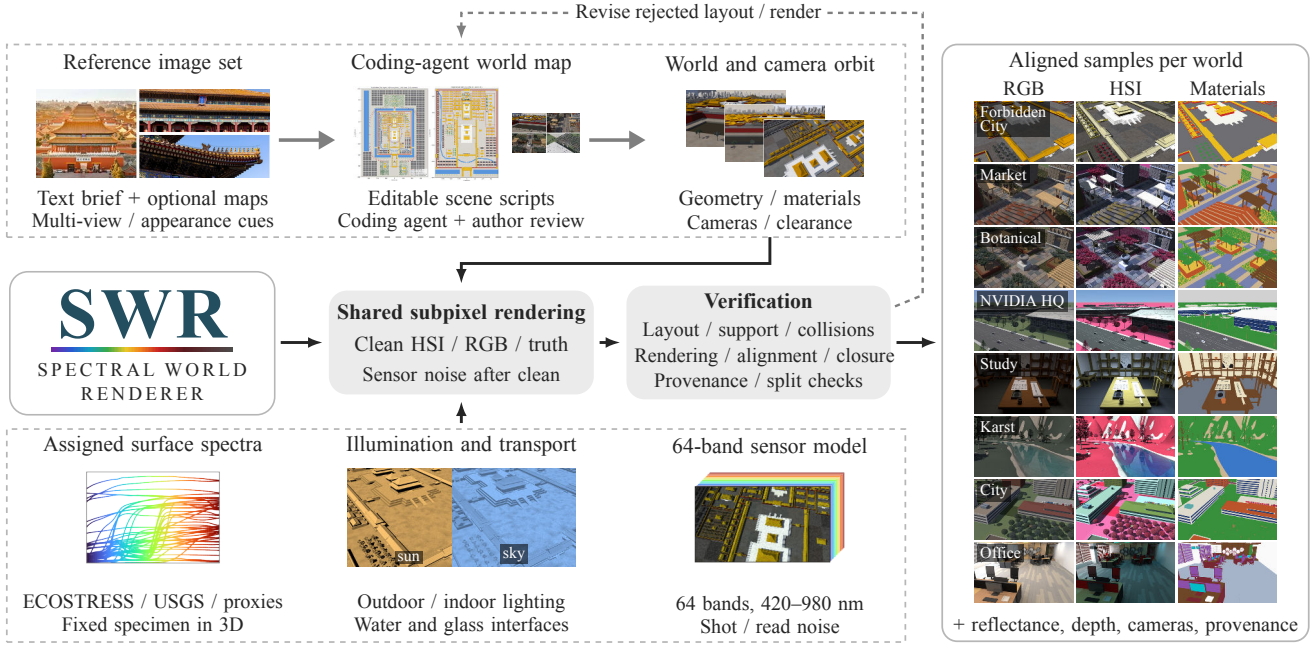


Figure 2. **The SWR world generator.** From reference photographs and a short brief, a coding agent writes an executable world whose every surface carries a library-measured spectrum. One shared sub-pixel rendering produces the hyperspectral and RGB observations together with the material of every pixel, so the ground truth is exact for the rendered world. Checks on layout and rendering return flawed worlds for revision, yielding aligned, verifiable samples across all eight worlds (right).

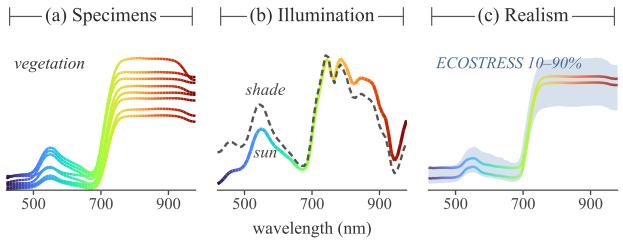


Figure 3. **Spectral variability within a material.** Grass specimens differ (a), and the same grass changes spectral shape between sunlight and shade (b). Both broadleaf spectra shown lie within independent leaf measurements (c).

spectral-variability and nonlinear models [27, 28]. These 2D abundance maps serve remote sensing and material analysis [20, 65]. UnMix-NeRF brings unmixing into a radiance field [63]. Like it, SWR learns endmembers and abundances jointly, but on explicit Gaussians and before compositing.

Programmatic world generation. Procedural rules [68, 69], language-model layouts [16, 88] and scene code [31, 48] build 3D worlds. Programmable and generative world renderers turn scene state or G-buffers into video [33, 34, 51]. None models spectra or proportions; the SWR world generator binds measured spectra to inspectable geometry, sampling images with their composition.

Synthetic benchmarks and spectral simulation. Synthetic and scanned datasets supply geometry, semantics or intrinsic images [23, 73, 78]; game and agent benchmarks add G-buffers and checkable scene code [19, 34, 74, 90]. Spectral renderers and remote-sensing simulators trace light per wavelength [21, 36, 64, 86], and ray-traced vegetation scenes give known abundances for 2D unmixing [13]. As Table 1 summarizes, none pairs multi-view spectra with area proportions; SWR-Bench samples both from the same sub-pixel points, varies specimens and illumination, and stores coefficients so every target can be regenerated.

3. Spectral World Renderer

Spectral World Renderer unifies 3D reconstruction and unmixing in one verifiable framework. The world generator and SWR-Bench (Sec. 3.1) render each image and its ground truth from the same camera samples, with indoor and outdoor worlds, labels and sub-pixel proportions and measured spectra. The renderer (Sec. 3.2) unmixes at every ray intersection, so one render yields spectra, abundances and material queries at any band count; it uses band-independent features, decoding before compositing and a separate intensity with multi-view appearance regularization (Sec. 3.3). Training uses no material labels; validation labels only select the number of endmembers K.

A scene is observed from V training cameras. Camera v provides RGB $\mathbf{I}_v \in [0,1]^{H\times W\times 3}$ and a co-registered hyperspectral cube $\mathbf{H}_v \in \mathbb{R}^{H\times W\times L}$ with L bands. The ray $\mathbf{r} = \mathbf{r}_v(p)$ of pixel p indexes its observed values $\mathbf{I}(\mathbf{r}) \in \mathbb{R}^3$ and $\mathbf{H}(\mathbf{r}) \in \mathbb{R}^L$. At a novel camera, it renders a spectrum $\hat{\mathbf{H}}(\mathbf{r})$ and abundances $\hat{\mathbf{A}}(\mathbf{r})$ over shared endmembers.

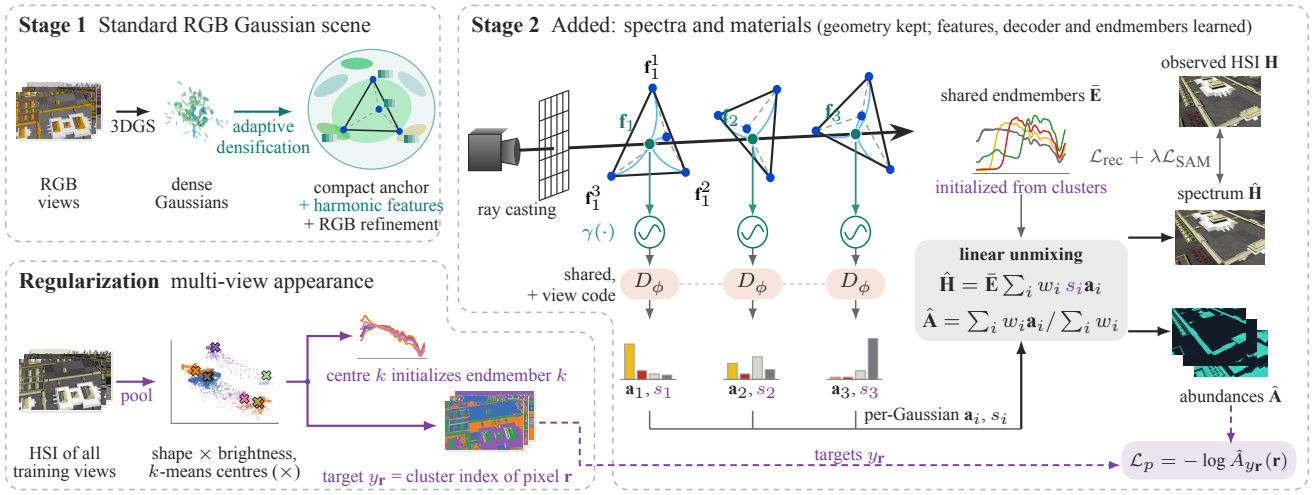


Figure 4. **Overview of Spectral World Renderer.** Stage 1 builds an RGB Gaussian scene with a standard pipeline; Stage 2 keeps its geometry and adds spectra and materials. Each Gaussian carries harmonic features whose size does not depend on the band count. A shared decoder maps the feature at each intersection to abundances and an intensity, which mix scene-wide endmembers; the same visibility weights render the spectrum and the abundances. Multi-view appearance clusters initialize and regularize the endmembers.

3.1. World Generator and SWR-Bench

Scoring an unmixing needs the material proportions of every pixel, which real captures cannot provide. The SWR world generator therefore renders images, labels and area proportions from the same camera samples (Figure 2).

Reference-guided world generation. Users ground each world in reference images, available maps and a text brief. A coding agent, the scene-building workflow for SWR-Bench rather than an evaluated system, turns them into executable scene programs: parameterized geometry and placement rules encode the layout, while materials, lighting and camera paths stay editable. The agent and the authors review previews for support, collisions and clearance.

Shared-sample ground truth. Images and labels rendered separately disagree at silhouettes and mixed pixels, so both use the same S sub-pixel camera samples: for the pixel indexed by ray $\mathbf{r}$, $\mathbf{x}_n$ is the first surface hit of sample n and $m(\mathbf{x}_n)$ its material class. A Lambertian specimen j with measured reflectance $\rho_j(\lambda)$ at wavelength λ has, under direct or sky illumination u, band radiance

$$v^u_{j,\ell} = \int \psi_\ell(\lambda)\, \rho_j(\lambda) \Phi_u(\lambda)/\pi \, \mathrm{d}\lambda, \quad u \in \{\mathrm{d}, \mathrm{s}\}, \qquad (1)$$

where ψ_ℓ is the response of band ℓ and $\Phi_\mathrm{d}, \Phi_\mathrm{s}$ are direct and sky irradiance. The vectors $\mathbf{v}^u_j \in \mathbb{R}^L$ and accumulated shading coefficients $C^u_j(\mathbf{r})$ give

$$\begin{aligned} \mathbf{H}_{\text{clean}}(\mathbf{r}) &= \textstyle\sum_j (C^\mathrm{d}_j \mathbf{v}^\mathrm{d}_j + C^\mathrm{s}_j \mathbf{v}^\mathrm{s}_j), \\ A_c(\mathbf{r}) &= \tfrac{1}{S} \textstyle\sum_{n=1}^{S} [m(\mathbf{x}_n) = c]. \end{aligned} \qquad (2)$$

Radiance and area proportions thus share one footprint; released proportions renormalize A_c over non-sky samples. Here c indexes ground-truth classes and k, below, learned endmembers. Transport grows across the worlds: four outdoor worlds follow this two-term sun/sky Lambertian model; NVIDIA HQ adds rough specular highlights, a glass reflection and a hazy sky; Karst water adds reflection, refraction and absorption; the Study and Office are path traced in Mitsuba [36]. Each effect is stored separately.

Measured spectra and their variation. Reflectance comes from ECOSTRESS and USGS measurements [42, 53], and every assignment is recorded, so each observation traces back to its spectra. No single spectrum describes a whole material: specimens differ within a class, and sunlight and skylight change the spectral shape of one surface, as Figure 3(a,b) shows. Each surface keeps one specimen across views. Outdoor worlds use ASTM G173 sunlight and skylight [1], interiors white sources [11]. We render 64 bands from 420 to 980 nm, clean and with shot and read noise [18].

Verification. Each world is checked for label alignment, spectral provenance and split separation: stored coefficients and contributions re-compose clean outdoor radiance, indoor transport is checked before noise, and assigned spectra lie within library measurements (Figure 3(c)).

3.2. Unmixing at Every Ray Intersection

After compositing, a rendered mixture can no longer be attributed to the Gaussians along the ray. Spectral World Renderer instead unmixes inside rendering: at every ray intersection, a shared decoder turns the local feature into abundances and an intensity, and compositing accumulates both, so one render yields spectra and abundances (Figure 4).

RGB anchor and harmonic features. Hyperspectral observations are noisy and vary with illumination, which across many views degrades the recovered geometry. We therefore fit the geometry as a 3DGS anchor on co-registered RGB images, supplied with the data or derived from visible-band responses; advances in RGB Gaussian splatting thus transfer directly. FastGS densification and

pruning [71] keep only primitives supported across views, so fewer features and intersections remain to render, optimize and unmix. Unlike HyperGS [82], the anchor requires no spectral autoencoder. It has N Gaussians G_i with mean $\boldsymbol{\mu}_i$, rotation $\mathbf{R}_i$, scale $\boldsymbol{\sigma}_i$ and opacity o_i; after fitting, we drop colours and fold the rasterizer's screen-space variance into the scales. Each Gaussian instead carries a neural harmonic texture [12], whose within-primitive variation lets few Gaussians represent fine detail: four feature vectors $\mathbf{f}_{im} \in \mathbb{R}^d$, $m = 1, \dots, 4$, with $d = 12$. A ray $\mathbf{r}$ queries G_i at its point of maximal response, whose tetrahedral barycentric coordinates $\beta_{im}(\mathbf{r})$ interpolate the vertex features:

$$\mathbf{f}_i(\mathbf{r}) = \sum_{m=1}^{4} \beta_{im}(\mathbf{r}) \mathbf{f}_{im} \in \mathbb{R}^d. \tag{3}$$

Different rays through one Gaussian thus read different features. We refine features and geometry with the NHT RGB shader, without further densification, and then freeze the geometry. Each Gaussian stores 11 geometry and 48 feature values, so its size does not grow with L.

Pre-compositing material decoder. Each intersection is decoded before compositing blends it with others. Its feature (Eq. 3), encoded as $\gamma(\mathbf{f}) = [\sin \mathbf{f}; \cos \mathbf{f}]$ [80], enters a shared MLP D_ϕ with a directional code $\mathbf{Y}(\mathbf{d}_i(\mathbf{r})) \in \mathbb{R}^9$: real spherical harmonics of degree ≤ 2 [70], where $\mathbf{d}_i$ points from the Gaussian centre to the camera. The decoder produces

$$\mathbf{z}_i(\mathbf{r}) = D_\phi(\gamma(\mathbf{f}_i(\mathbf{r})), \mathbf{Y}(\mathbf{d}_i(\mathbf{r}))) \in \mathbb{R}^{K+1}. \tag{4}$$

Here K is the number of shared endmembers. The first K outputs yield abundances $\mathbf{a}_i(\mathbf{r}) = \mathrm{softmax}(\mathbf{z}_i^{1:K}(\mathbf{r}))$; the last gives intensity $s_i(\mathbf{r}) = \mathrm{softplus}(z_i^{K+1}(\mathbf{r}))$. The MLP has three width-128 SiLU hidden layers; a zero-initialized linear map adds the directional code to layer one.

These coefficients mix a scene-wide dictionary $\bar{\mathbf{E}} = [\bar{\mathbf{e}}_1, \dots, \bar{\mathbf{e}}_K] \in \mathbb{R}^{L \times K}$. Each endmember is parameterized by $\boldsymbol{\eta}_k \in \mathbb{R}^L$ as

$$\bar{\mathbf{e}}_k = \frac{\mathrm{sigmoid}(\boldsymbol{\eta}_k)}{\|\mathrm{sigmoid}(\boldsymbol{\eta}_k)\|_2} \in \mathbb{R}^L_{\geq 0}. \tag{5}$$

Normalized endmembers carry shape and s_i magnitude; each intersection spectrum is a scaled linear mixture [5]:

$$\mathbf{h}_i(\mathbf{r}) = s_i(\mathbf{r}) \bar{\mathbf{E}} \mathbf{a}_i(\mathbf{r}) \in \mathbb{R}^L_{\geq 0}. \tag{6}$$

Sec. 3.3 initializes and regularizes the dictionary.

Joint spectral and mixture rendering. Spectra and abundances should describe the same visible surfaces. With intersected Gaussians sorted front to back, with opacity $\alpha_i(\mathbf{r})$, transmittance $T_i(\mathbf{r}) = \prod_{i'<i}(1 - \alpha_{i'}(\mathbf{r}))$ gives visibility weights $w_i(\mathbf{r}) = T_i(\mathbf{r})\alpha_i(\mathbf{r})$ [37], which render both:

$$\hat{\mathbf{H}}(\mathbf{r}) = \sum_i w_i(\mathbf{r}) \mathbf{h}_i(\mathbf{r}), \quad \hat{\mathbf{A}}(\mathbf{r}) = \frac{\sum_i w_i(\mathbf{r}) \mathbf{a}_i(\mathbf{r})}{\sum_i w_i(\mathbf{r})}. \tag{7}$$

Each $\mathbf{a}_i$ is non-negative and sums to one [26], and for a non-empty ray $\hat{\mathbf{A}}$ is a convex combination of them, so it is too. Intensity scales $\hat{\mathbf{H}}$ but not $\hat{\mathbf{A}}$: $\hat{\mathbf{H}} = \bar{\mathbf{E}} \sum_i w_i s_i \mathbf{a}_i$, whereas $\hat{\mathbf{A}}$ weights each Gaussian by visibility w_i alone, giving endmember mixing weights. Because decoding precedes accumulation, each intersection keeps its own mixture; averaged over its training rays, each Gaussian's abundance can be queried in 3D (Sec. 4.4).

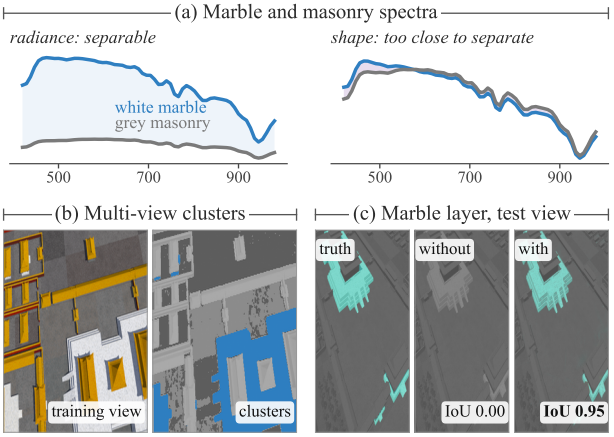


Figure 5. **Multi-view appearance clusters** (Forbidden City). (a) White marble is four times brighter than grey masonry but only 3.8° apart in spectral shape; since intensity absorbs brightness, one endmember can explain both. (b) Clustering pixels of all training views by shape and brightness separates them, with clusters shared by every view. (c) As a weak regularizer during training, these clusters keep marble apart from masonry.

3.3. Multi-View Appearance Regularization

Assignment ambiguity. Accurate spectra do not imply a correct decomposition. Eq. 6 fits each spectrum but does not fix which endmember explains it. Two similar materials may share one endmember while the intensity s absorbs their brightness difference, like the scaling factors of extended mixing models [14, 28]; the spectra are reconstructed well, yet the materials merge in $\hat{\mathbf{A}}$. We therefore derive an initial endmember for each appearance from the multi-view training data and keep this assignment as a weak regularizer, as Figure 5 illustrates.

Over-complete dictionary. One material spans several spectra (Figure 3). Following endmember bundles and multiple-endmember unmixing [3, 72, 77], we therefore choose K larger than the number of materials, one endmember per appearance mode. Extra endmembers alone do not decide which one serves which surface; without guidance they drift to dominant materials, as Figure 5(c) shows.

Multi-view appearance clusters. Without labels, we pool foreground pixels of all training views and describe each 3×3-smoothed spectrum $\tilde{\mathbf{H}}(\mathbf{r})$ by its shape and brightness:

$$\mathbf{q}_\mathbf{r} = \left[\frac{\tilde{\mathbf{H}}(\mathbf{r})}{\|\tilde{\mathbf{H}}(\mathbf{r})\|_2}, \; \kappa\, b(\tilde{\mathbf{H}}(\mathbf{r})) \right], \tag{8}$$

where b is the log norm of the spectrum, standardized robustly over the training views and clipped. The bounded

Table 2. **Material identification on SWR-Bench.** Spectral World Renderer has the highest mIoU in every world, the best or second-best PSNR and SAM, and learns the endmembers closest to the true spectra; a lower spectral angle (3DGS in Market and Study) does not imply better identification. Rare: classes under 3 % of test pixels; Spectra: specimens; EM in degrees.

					UnMix-NeRF [63]				3DGS [37] + VCA / NNLS				Spectral World Renderer			
Scene	Setting	Classes	Rare	Spectra	PSNR↑	SAM↓	mIoU↑	EM↓	PSNR↑	SAM↓	mIoU↑	EM↓	PSNR↑	SAM↓	mIoU↑	EM↓
Forbidden City	outdoor	9	3	47	26.13	7.37	0.19	27.4	25.03	5.05	0.17	30.7	28.37	4.45	0.55	4.4
Market	outdoor	6	0	29	31.87	3.61	0.13	30.9	30.02	1.85	0.24	26.9	33.63	1.95	0.78	2.4
Botanical	outdoor	6	0	29	30.24	5.99	0.10	38.8	29.57	3.12	0.22	27.1	32.88	2.97	0.71	2.4
NVIDIA HQ	outdoor	8	2	27	18.29	14.98	0.08	43.0	22.27	12.31	0.13	24.4	25.30	8.25	0.31	3.7
Study	indoor	5	2	5	33.78	2.81	0.20	22.9	32.53	1.52	0.30	14.9	33.14	2.01	0.52	0.9
Karst	outdoor	5	2	10	26.91	3.29	0.38	25.8	27.19	2.26	0.35	28.8	28.10	2.15	0.73	2.9
City	outdoor	11	5	56	27.00	4.40	0.10	30.4	25.61	4.21	0.21	29.1	28.18	3.53	0.52	3.1
Office	indoor	8	3	35	25.33	6.65	0.11	18.8	25.01	5.03	0.17	41.3	27.54	4.31	0.53	3.3

weight κ keeps brightness secondary to shape, yet lets it separate nearly parallel spectra that the spectral angle alone would merge. Farthest-point seeding and k-means group the descriptors into K clusters; as in clustering-based self-supervision [7], each pixel's cluster becomes its pseudo-label $y_{\mathbf{r}}$, and cluster centre k initializes endmember k.

Persistent assignment regularization. During training, each pseudo-labelled pixel should concentrate its abundance (Eq. 7) on the endmember of its cluster:

$$\mathcal{L}_p = -\frac{1}{|\mathcal{B}_p|} \sum_{\mathbf{r}\in\mathcal{B}_p} \log \hat{A}_{y_{\mathbf{r}}}(\mathbf{r}), \tag{9}$$

where $\hat{A}_{y_{\mathbf{r}}}(\mathbf{r})$ is entry $y_{\mathbf{r}}$ of $\hat{\mathbf{A}}(\mathbf{r})$ and $\mathcal{B}_p$ holds the pseudo-labelled pixels of the batch whose rays hit the scene. Because a Gaussian appears in many training views, its abundances must agree with the pseudo-labels of all of them. Like semantic lifting in radiance fields [75, 89, 94], the loss thus lifts 2D appearance clusters into a 3D assignment by multi-view agreement. It constrains which endmember explains a pixel, not its spectrum or intensity. Endmembers stay frozen for 3k updates while the weight of $\mathcal{L}_p$ decays (Sec. 3.4), so distinct appearances do not merge.

3.4. Optimization

Stage 1 fits the anchor and refines geometry and features on RGB with $\mathcal{L}_{\text{RGB}} = 0.8\mathcal{L}_1 + 0.2\mathcal{L}_{\text{D-SSIM}}$. Stage 2 freezes the geometry and updates features, decoder and endmembers on a uniform pixel batch $\mathcal{B}$ from one training camera, with

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{rec}} + \lambda_{\text{SAM}}\mathcal{L}_{\text{SAM}} + \lambda_p(t)\mathcal{L}_p, \\ \mathcal{L}_{\text{rec}} &= \tfrac{1}{|\mathcal{B}|L} \textstyle\sum_{\mathcal{B}} \|\hat{\mathbf{H}}_{\mathbf{r}} - \mathbf{H}_{\mathbf{r}}\|_1, \\ \mathcal{L}_{\text{SAM}} &= \tfrac{1}{|\mathcal{B}_+|} \textstyle\sum_{\mathcal{B}_+} \text{SAM}(\hat{\mathbf{H}}_{\mathbf{r}}, \mathbf{H}_{\mathbf{r}}), \end{aligned} \tag{10}$$

where $\mathcal{B}_+$ has non-zero targets, SAM is in radians, $\lambda_{\text{SAM}} = .05$ and $\lambda_p(t) = \max(.005, .05(1 - t/3000))$ at update t.

4. Experiments

Data and baselines. SWR-Bench has eight indoor and outdoor worlds of known composition. For materials, we compare UnMix-NeRF [63] and an L-channel 3DGS [37] unmixed afterwards with VCA [59] endmembers from training-view renders and NNLS [45]. For fidelity, we add HyperGS [82] and DD-HGS [58]. Real data span 21 to 204 bands: four multispectral NeSpoF scenes [40], the 204-band HSI-SC apple [44] and five HS-NeRF scenes [9].

Table 3. **Effect of RGB feature refinement.** Refining geometry and harmonic features on the RGB anchor gives the decoder spatially structured rather than random features.

	Market (SWR-Bench)			HSI-SC apple		City
	mIoU↑	PSNR↑	SAM↓	PSNR↑	SAM↓	train↓
w/o refinement	0.66	30.73	3.20	28.44	1.67	–
SWR	**0.78**	**33.63**	**1.95**	**30.99**	**1.34**	6 min 37 s

Protocol. Methods train on noisy 480×270 views; PSNR and the spectral angle SAM [43] (mean and 95th percentile, in degrees) use clean 960×540 foreground truth. For mIoU, each endmember maps to the class of most training pixels it dominates; with this map frozen, each test pixel takes the class of its dominant endmember. EM is the mean angle between each class's mean pure-pixel spectrum and its endmember, paired by minimum total angle (Appendix A).

Implementation details. Table S4 lists the training settings of all methods. Validation mIoU selects the endmember count K per world from the class count, 16, 24 and 32: K=24 for Market, NVIDIA HQ and Study, K=32 elsewhere. Real captures use K=16; baselines use the class count. At K=24, 3DGS+VCA/NNLS stays below SWR in every world but Study, where the two tie (Table S5).

4.1. Material Identification

Identification. In Table 2, SWR reaches a mean mIoU of 0.58, against 0.16 and 0.22 for UnMix-NeRF and 3DGS+VCA/NNLS, and its endmembers lie 0.9–4.4° from the class spectra, against at least 14.9° for the baselines.

Fidelity–identification gap. A lower SAM does not imply better labels: 3DGS reaches a lower SAM yet a far lower mIoU on Market and Study (Table 2, Figure 6). Noise and the learned decomposition both limit identification: NNLS with all measured specimens reaches 0.84–0.94 mIoU on clean radiance but 0.32–0.77 under noise (Appendix A).

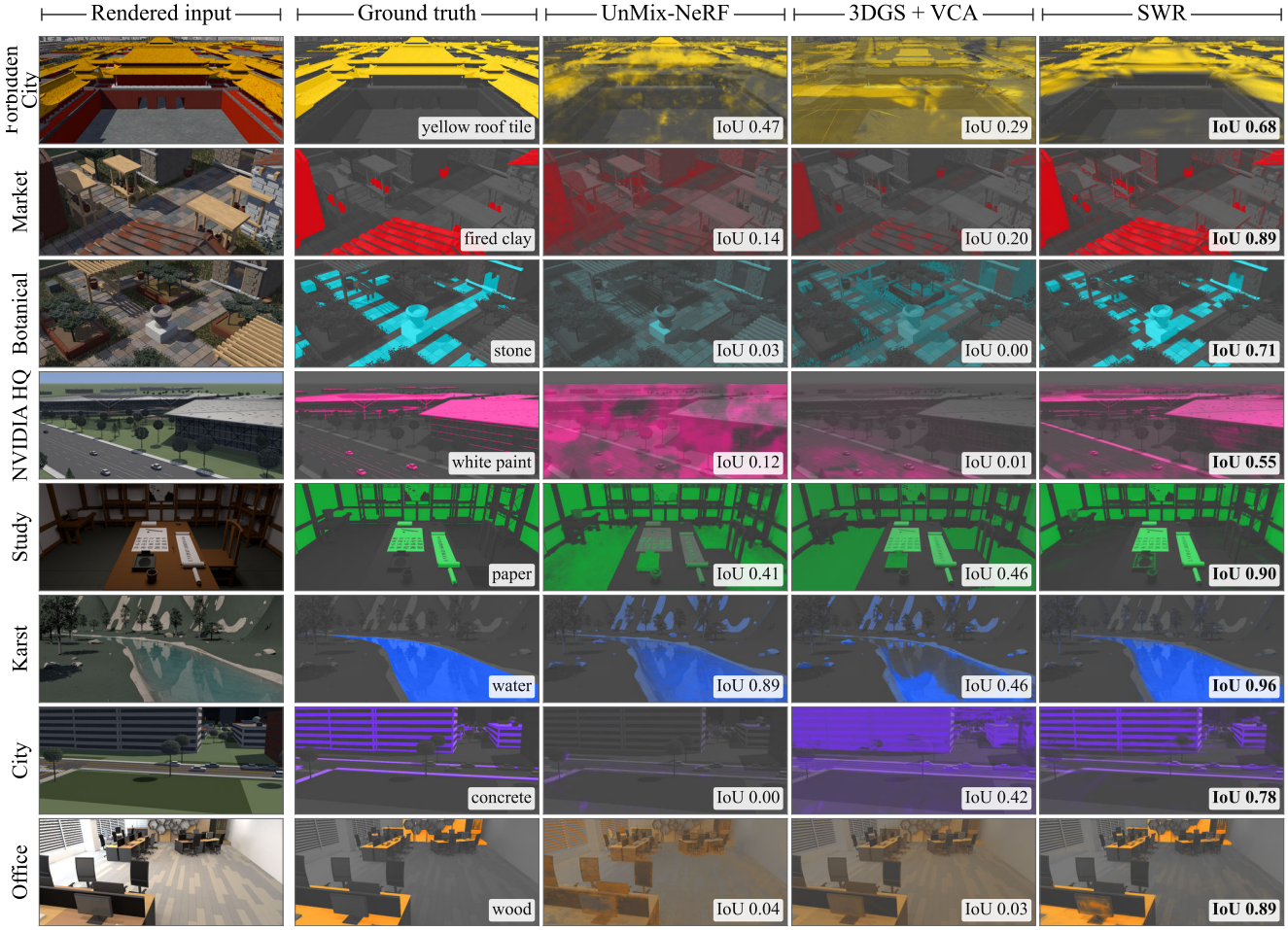


Figure 6. **Material layers on SWR-Bench.** Across all eight worlds, each row tints one material by its abundance in a test view, for a material to which every method maps an endmember; IoU: summed abundance > 0.5 against the true mask. Unmixing rendered spectra afterwards (3DGS + VCA) or inside a radiance field (UnMix-NeRF) spreads the material over unrelated surfaces. All rows use the models of Table 2.

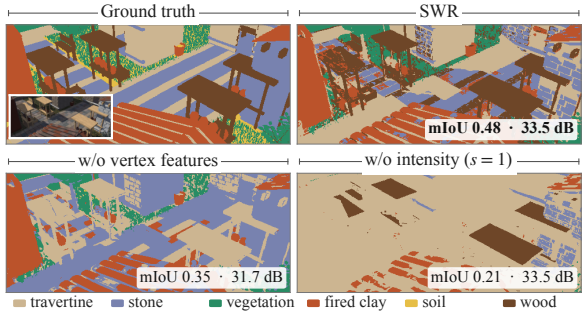


Figure 7. **Decoder variants on Market** (test view, no $\mathcal{L}_p$): material maps with mIoU and PSNR for this view; inset: input.

4.2. Ablation and Analysis

RGB feature stage. Refinement on the fixed anchor is cheap: it takes 36.5 s of the 397 s City pipeline, yet refining geometry and feature layout improves Market by 2.9 dB and 0.12 mIoU and the real apple by 2.6 dB, as Table 3 shows.

Within-primitive variation. A Gaussian crossing an edge or shadow needs more than one mixture, as Figure 7 tests on Market without $\mathcal{L}_p$. One feature per Gaussian costs 1.6 dB and lowers mIoU from 0.47 to 0.34. Fixed intensity keeps PSNR but lowers mIoU to 0.22, as mixtures absorb shading.

Unmixing inside rendering. Decoding each intersection gives every Gaussian its own abundances, which the queries of Sec. 4.4 select. Without regularization, decoding after compositing raises Market PSNR from 33.72 to 34.37 dB yet lowers mIoU from 0.47 to 0.37 as ray-averaged features blend surfaces; with it, per-intersection decoding identifies best in every tested world (Table S6).

Material allocation limits. Appearance regularization recovers rare Forbidden City marble (Figure 5(c)) but misses ink in Study; NNLS with all measured specimens separates most rare classes SWR merges (Figure S1).

4.3. Real Captures and Efficiency

Fidelity without material truth. SWR has the highest PSNR on each real dataset (Table 5). Its margin peaks on the 204-band apple (Figure 8), beating HyperGS on all metrics. On HS-NeRF, 3DGS has a lower mean and p95 angle.

Table 4. **Training and rendering cost.** FPS: views/s with spectra and label maps. Ours: our optimized fused fp16 decoder, skipping hits of weight $< 10^{-4}$ (99.97 % identical labels); baselines: reference code. City training spans all four stages, anchor to decoder.

Method	City train↓	FPS↑ at $960 \times 540 \times 64$ Market	Bot.	Study	Karst	Mean
UnMix-NeRF	24 min 17 s	0.18	0.21	0.21	0.12	0.18
3DGS+VCA/NNLS	85 min 2 s	2.53	2.51	4.70	4.73	3.6
SWR	**6 min 37 s**	**36.5**	**39.3**	**36.3**	**27.4**	**34.9**

Table 5. **Spectral reconstruction on real captures.** L: bands; p95: 95th-percentile SAM; mean of 4 NeSpoF / 5 HS-NeRF scenes; baseline: best of 3DGS, HyperGS, DD-HGS.

Source	L	PSNR-leading baseline Method	PSNR↑	SAM↓	p95↓	SWR PSNR↑	SAM↓	p95↓
NeSpoF	21	HyperGS	28.36	4.58	9.44	**29.39**	**3.86**	**8.04**
Apple	204	HyperGS	27.95	1.64	4.88	**30.67**	**1.35**	**3.87**
HS-NeRF	128 141	3DGS	26.32	**5.71**	**16.75**	**26.51**	5.97	17.31

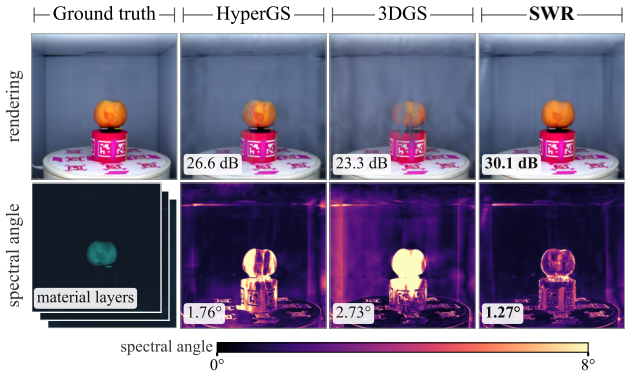


Figure 8. **HSI-SC apple, test view.** Rendering with PSNR (top) and per-pixel spectral angle with its mean (bottom); stacked inset: SWR material layers.

Efficiency. SWR is fastest in Table 4: City trains in 6 min 37 s versus 24 min for UnMix-NeRF and 85 min for 3DGS+VCA/NNLS, and it renders 34.9 views/s.

4.4. Application: Material Queries

Seeing beyond RGB. Leaves have a red edge near 700 nm: chlorophyll absorbs red light and leaf tissue reflects near-infrared. Minerals and dry leaves lack it, so foliage-coloured mineral leaves (2° apart in RGB, 42° in spectrum) and dry crowns get their own endmembers (Figure 9(b,c)).

Querying primitives directly. Hyperspectral images usually isolate such materials with supervised classifiers [10, 29, 54, 81] or subspace clustering [92], which need labels or a run per image. SWR needs neither: abundances live on explicit, rasterizable Gaussians, so a material query selects primitives, consistently across views. The mineral-leaf query retrieves no ordinary tree, whereas the RGB query retrieves eleven, as Figure 9(a) shows. A measured spectrum can also be the query: with the world's known sun and sky

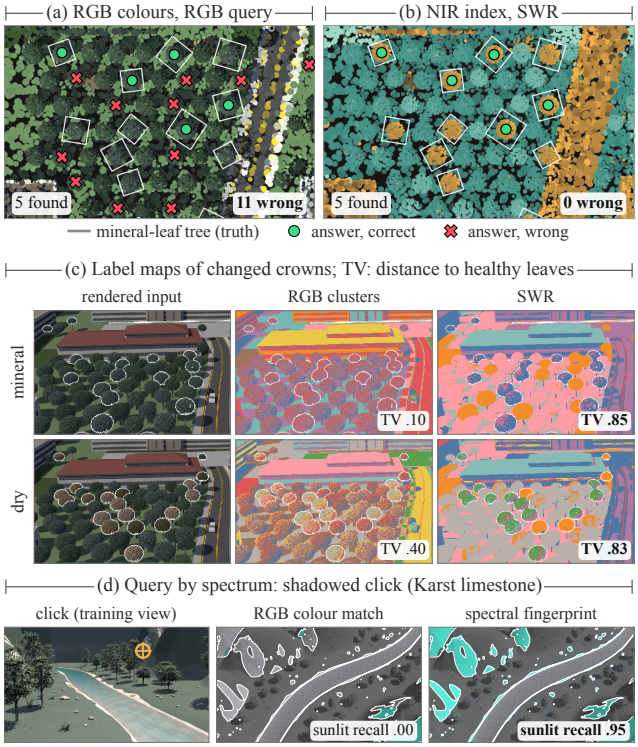


Figure 9. **Material queries.** In City, foliage-coloured mineral leaves replace the boxed crowns. (a) The same query on RGB also retrieves ordinary trees. (b) Lacking the NIR rise of foliage, the mineral is retrieved alone. Counts cover the instances returned by the type query (Table S11). (c) SWR separates mineral and dry crowns from healthy foliage; RGB clusters mix them. (d) A spectrum clicked on shadowed limestone (Karst) also retrieves the sunlit rock, which RGB misses (recall over all test views).

irradiance factored out, a pixel clicked in shadow retrieves 77 % of the sunlit part of its material (Figure 9(d); four outdoor worlds, 3 ms per query; Appendix H).

5. Conclusion

Spectral World Renderer brings hyperspectral unmixing into 3D as one verifiable framework: its world generator renders images with their exact material composition, and its renderer unmixes at every ray intersection, so one render yields spectra and material maps checkable against known truth. Trained without material labels, SWR identifies materials best in all eight worlds and has the best mean PSNR on real captures; it trains City in under seven minutes, renders 34.9 labelled views/s, and its per-Gaussian abundances answer queries by click, reference spectrum, band or a name attached through a training-view map, even for same-coloured materials. Future work will enrich light transport and sensor models, allocate endmembers to rare materials and handle dynamic scenes.

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Supplementary Material

Contents. Figures, tables and equations carry the prefix S; unprefixed numbers refer to the main paper.

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A. SWR-Bench Worlds and Protocol

Generation. A coding agent builds each world from reference photographs, available maps and a written brief, revising it from rendered previews; the authors review the previews and approve each world, and the worlds, scene programs and code will be released (Table S1). Each surface keeps one measured specimen in world space across all views.

Shared samples. Spectral images, RGB and ground truth share 16 sub-pixel samples per pixel, so labels and area proportions describe the first-hit surfaces. Radiance reflected or transmitted by water and glazing is therefore attributed to the water or glazing class, and interreflected light to the surface on which it is observed; transmitted and interreflected light is stored as separate contributions.

Rendering. All worlds share 64 Gaussian bands centred from 420 to 980 nm (20 nm full width at half maximum) at 960×540. Outdoor worlds use Lambertian surfaces under ASTM G173 sunlight and skylight [1]; NVIDIA HQ adds glossy highlights, glass reflection and an analytic sky, and Karst adds refracting water with pure-water absorption. The Study and Office are path traced in Mitsuba [36] with interreflection, sampling reflectance at the band centres. RGB is noiseless: CIE 1931 colour matching gives XYZ, which is adapted to D65 by the Bradford transform (70 % partial adaptation indoors) and converted to sRGB with one fixed exposure per world.

Noise. Every world is released clean and noisy, with Poisson and read noise added per band to the 960×540 radiance L. With N_e electrons at the reference radiance L_{ref} and read noise σ_r (Table S1), the observation is $\tilde{L} = (L_{\mathrm{ref}}/N_e)\,[\mathrm{Poisson}(N_e L/L_{\mathrm{ref}}) + \mathcal{N}(0, \sigma_r^2)]$. Outdoors, L_{ref} is the maximum clean radiance over all training pixels and bands, and negative values are clipped to zero; in the Study and Office, it is the 99.5th percentile of the first training view, excluding emitters, without clipping.

Protocol. Training views at 480×270 are 2×2 box averages of the noisy cubes divided by L_{ref}; evaluation uses the native 960×540 views. All methods share the splits, classes and stereo point cloud. Each world has full primary-ray coverage, at least 99.9 % agreement between instance and material maps, and outdoor radiance that re-composes exactly from the stored coefficients.

Classes and City variants. Finer labels are merged into one class when they play the same material role with similar spectra, for example the marbles and granite of Market and Botanical, the near-neutral grey paving and brick of Forbidden City, and bark and soil; spectrally distinct labels remain separate even when rare, such as painted eaves and yellow road paint, whereas white road paint (0.3 % of City pixels) is merged into asphalt. Four City variants change one factor each: foliage-coloured mineral leaves (*metamer*), dry leaves on 87 trees (*stress*), and a morning or afternoon sun. Every object keeps its instance and spectral sample.

Metrics. PSNR and spectral angle are computed against the clean radiance on the ground-truth support.

- *Labels.* Endmembers are mapped many-to-one to classes by majority vote on training views. With the map π frozen, each test pixel takes the class of its dominant endmember, $c = \pi(\arg\max_k \hat{A}_k)$, not the class of largest summed abundance; endmembers that dominate no training pixel stay unassigned.
- *Endmember angle (EM).* Each class is paired with a distinct endmember by the linear-sum (Hungarian) assignment of minimum total angle to the mean clean spectrum of its pure training pixels (area proportion at least 0.9). With $K \geq C$, the remaining $K - C$ endmembers do not enter EM, and the map used for mIoU plays no role.
- *True-spectrum reference (GT dict.).* Every second test pixel along each axis is unmixed by non-negative least squares against all measured specimens of the world, each under sunlight and skylight, and labelled by the largest class-summed contribution; Karst is scored on its opaque classes. The reference reaches 0.84–0.94 mIoU on clean radiance but 0.32–0.77 under noise, and gives the true-spectrum IoU of Figure S1. It is undefined for the path-traced Study and Office, which include interreflection, and for NVIDIA HQ, whose glass reflections and haze fall outside the first-surface dictionary.

Rare classes. Class IoU falls with class frequency, while the true spectra still separate most rare classes (Figure S1); the errors therefore lie in endmember allocation.

Additional views. Figures S10 and S11 show the simulated hyperspectral observations of one test view per world, and Figures S12–S15 three test views of each world with the ground truth and the Spectral World Renderer prediction, using the models of the main table and their frozen training-view maps. Views are chosen by camera separation from a fixed reference camera, not by prediction scores.

Table S1. **The eight worlds of SWR-Bench.** Classes and spectra: material classes and distinct measured spectra (Table S2). Views are split by camera into training, validation and test. Noise: electrons N_e at the reference radiance and read noise σ_r. Obs.: PSNR of the noisy test views against the clean ones. City variants keep the City cameras and splits. Bold: totals.

						Views				
World	Content	Cameras	Transport	Classes	Spectra	Train	Val	Test	N_e / σ_r	Obs. (dB)
Forbidden City	palace: golden roofs, courtyards, water	3 orbits, 45–430 m	two-term	9	47	135	21	24	5,000 / 2	45.4
Market	market square: stalls, pottery, walls	3 rings, inward and downward	two-term	6	29	135	21	24	5,000 / 2	45.6
Botanical	botanical garden: planters, trees, pergolas	3 rings, inward and downward	two-term	6	29	135	21	24	5,000 / 2	46.4
NVIDIA HQ	offices: glazed facades, 300 trees	spiral, facade, campus routes	extended	8	27	180	28	32	5,000 / 2	46.7
Study	scholar's study: shelves, paper, ink, ceramics	eye-level and elevated orbits	path traced	5	5	64	16	16	3,000 / 3	44.8
Karst	river valley: limestone cliffs, clear water	river banks, cliff, high loop	two-term + water	5	10	128	32	32	5,000 / 2	44.3
City	campus block: roads, 297 trees, vehicles	4 orbits, 16–95 m	two-term	11	56	180	28	32	5,000 / 2	45.2
Office	office: 35 LED downlights, daylight	eye-level and elevated loops	path traced	8	35	64	16	16	3,000 / 3	46.0
Total					**125**	**1,021**	**183**	**200**		
City variants	metamer, stress, morning, afternoon	as City	two-term	11–12	56–60	as City			as City	–

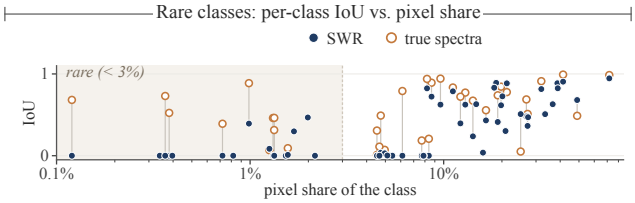


Figure S1. **Per-class IoU and class frequency.** Per-class IoU against pixel share over the eight worlds, with K equal to the class count. NNLS with all measured specimens (GT dict.) separates most rare classes that Spectral World Renderer merges.

B. Spectra of SWR-Bench

Sources and selection. Reflectance comes from ECOSTRESS [53] and the USGS Spectral Library 7 [42] (Table S2). A record is accepted only if it is a reflectance measurement that covers 400–1000 nm with gaps of at most 20 nm and values in $[0, 1]$, and if its metadata satisfy a written rule for its class, for example photosynthetic grass but no dry grass, or bark only on trunks. Accepted records are interpolated on a 1 nm grid without smoothing or extrapolation and integrated against 64 Gaussian bands of 20 nm full width at half maximum, truncated at ± 20 nm. Several specimens per class and separate spectra under sunlight and skylight give each material a range of appearances (Figure 3 of the main paper).

Realism. Rendered natural materials agree closely with independent library measurements of the same type, excluding the rendered records themselves (Table S3): 37 of 40 types lie within 3° of a reference, and all 21 vegetation spectra pass red-edge and NDVI checks [30, 83]. The dry and foliage-coloured substitutes of the City variants fail these checks, as designed. Man-made materials are more varied, and pigment proxies for paints deviate most.

Proxies. Some objects use a measured specimen of a related material, such as pigment powders for paints and road markings, cotton for paper, or granite for an inkstone.

C. Implementation Details

RGB anchor. Table S4 lists the training settings of all methods. The anchor uses multi-view density control [71], which clones and splits Gaussians that cover many high-error pixels and prunes by opacity and multi-view importance. After fitting, colours are discarded and the rasterizer's screen-space variance is folded into the scales,

$$\boldsymbol{\sigma}_i^2 \leftarrow \boldsymbol{\sigma}_i^2 + 0.3\,\mathrm{med}_v(z_{iv}/f_v)^2, \tag{S1}$$

where z_{iv} is the depth of Gaussian i in training camera v, f_v the focal length in pixels of the 480×270 training views, and the median runs over the training cameras that see the Gaussian; the term transfers the $0.3\,\mathrm{px}^2$ low-pass variance of the anchor rasterizer to the 3D kernel. The harmonic features are then refined for 5k steps without densification.

Vertex interpolation. The canonical tetrahedron is regular with inradius 1 in the frame of the Gaussian scaled by $\boldsymbol{\sigma}_i$, with vertices $(\pm\sqrt{6}, -\sqrt{2}, -1)$, $(0, 2\sqrt{2}, -1)$ and $(0, 0, 3)$. The barycentric coordinates of the point of maximal response are affine in its position and are not clamped, so they extrapolate linearly outside the tetrahedron.

Appearance clusters. Sampled pixel spectra $\mathbf{x}$ (Table S4) are described by $[\mathbf{u}, \kappa b]$, with the unit spectrum $\mathbf{u} = \mathbf{x}/\|\mathbf{x}\|$ and the standardized log norm $b = \mathrm{clip}\big((\log\|\mathbf{x}\| - m)/\max(0.1,\ 1.4826\,\mathrm{MAD}), -4, 4\big)$, where m and MAD are the median and the median absolute deviation of $\log\|\mathbf{x}\|$. The weight $\kappa = \mathrm{clip}(0.5\,\mathrm{med}\|\mathbf{u} - \bar{\mathbf{u}}\|, 0.05, 0.2)$, with $\bar{\mathbf{u}}$ the normalized mean of $\mathbf{u}$, adapts to each scene. Farthest-point seeding and 20 Lloyd steps on the unit sphere give K clusters. Endmember k is parametrized as $\mathbf{e}_k = \mathbf{s}(\boldsymbol{\eta}_k)/\|\mathbf{s}(\boldsymbol{\eta}_k)\|$ with the logistic function $\mathbf{s}$, and $\boldsymbol{\eta}_k = \mathrm{logit}(\mathbf{c}_k)$ is initialized from the unit spectral part $\mathbf{c}_k$ of cluster centre k, clipped to $[10^{-5}, 1 - 10^{-5}]$. On Market (K=24), the regularization and the cluster initialization each contribute: 0.771 and 0.732 mIoU alone, 0.784 together and 0.505 without either; the adaptive κ also outperforms fixed values of 0.05 and 0.2.

Table S2. **Material classes and measured spectra of the eight SWR-Bench worlds.** Class names describe material roles, which may use measured proxies. Counts deduplicate native files; $n_{\mathrm{E}}+n_{\mathrm{U}}$ denotes ECOSTRESS and USGS records, a single number ECOSTRESS, and a subscript U USGS. Brackets: number of finer labels before merging. Forbidden City uses pigment proxies for yellow tiles and some paints, measured reflectance on opaque water surfaces, and bright solid-material proxies for haze. Karst water uses pure-water absorption and a Fresnel interface instead of a reflectance endmember. City variants share the City geometry, cameras and sensor.

World	Classes	Spectra	E / U	Classes with their number of spectra	Illumination
Forbidden City	9 (11)	47	27 / 20	yellow glazed tile 5_{U}; vermilion paint 3_{U}; vegetation 8; white marble 4; grey masonry 4+5; painted eaves 2+2; soil/bark 7; water 1+2; distant haze 1+3	G173 sun + sky
Market	6 (8)	29	29 / 0	travertine 1; stone (white marble, pink marble, granite) 7; vegetation 8; fired clay 4; sandy soil 4; wood/bark 5	G173 sun + sky
Botanical	6 (8)	29	29 / 0	travertine 1; stone (white marble, pink marble, granite) 7; vegetation 8; fired clay 4; sandy soil 4; wood/bark 5	G173 sun + sky
NVIDIA HQ	8	27	21 / 6	white coating/marking 2+1; glazing 1; vegetation 8; dark panels/frames 2_{U}; concrete 1+2; asphalt 1+1; soil/bark 7; wood 1	G173 sun + sky
Study	5	5	3 / 2	aged wood 1; paper 1_{U}; ink 1_{U}; inkstone 1; ceramic 1	5000 K panel
Karst	5	10	10 / 0	limestone 2; vegetation 5; soil 1; wood/bark 2; water (absorption model)	G173 sun + sky
City	11 (14)	56	33 / 23	concrete/masonry 4+3; soil/bark 6; vegetation 6+1; fired clay 6+2; asphalt/markings 1+3; yellow road paint 2_{U}; dark glazing 2; painted facade 1+5; metal roofing 2+2; vehicle 4+4; dark backdrop 1+1	G173 sun + sky
Office	8	35	11 / 24	light oak 2+4; dark laminate 2+3; textile 5_{U}; grey vinyl 3_{U}; white paint/plastic 6_{U}; vegetation 4; metal 2+2; display 1+1	LED-B3 + G173
City-metamer	12 (15)	59	35 / 24	concrete/masonry 4+3; soil/bark 6; vegetation 6+1; fired clay 6+2; asphalt/markings 1+3; yellow road paint 2_{U}; dark glazing 2; painted facade 1+5; metal roofing 2+2; vehicle 4+4; dark backdrop 1+1; metameric substitute 2+1	as City
City-stress	11 (14)	60	36 / 24	concrete/masonry 4+3; soil/bark 6; vegetation 9+2; fired clay 6+2; asphalt/markings 1+3; yellow road paint 2_{U}; dark glazing 2; painted facade 1+5; metal roofing 2+2; vehicle 4+4; dark backdrop 1+1	as City

Table S3. **Reflectance checks per world,** over 420–980 nm. Spectra: distinct measured spectra rendered. Ref.: spectra with at least three independent library measurements of the recorded source type; $< 3^\circ$: how many are within 3° of one; med.: median nearest angle. Veg.: green-leaf feature checks. Missing independent references are unevaluated, including Forbidden City pigment-to-tile, opaque-water and distant-background proxies. Counts concern source measurements, not surveyed historical materials. A shared spectrum counts in each world; summaries count each (file,type) pair once (154 pairs of 125 distinct spectra). Natural: vegetation, bark, soil and rock.

World	Spectra	in $[0, 1]$	Ref.	$< 3^\circ$	med. ($^\circ$)	Veg.
Forbidden City	47	47/47	24	22	1.28	8/8
Market	29	29/29	21	17	2.01	8/8
Botanical	29	29/29	21	17	2.01	8/8
NVIDIA HQ	27	27/27	19	17	1.35	8/8
Study	5	5/5	2	1	6.09	–
Karst	10	10/10	9	9	0.60	5/5
City	56	56/56	30	20	2.07	7/7
Office	35	35/35	18	8	3.46	4/4
Six outdoor worlds	119	119/119	61	47	1.47	17/17
All eight worlds	154	154/154	78	55	2.00	21/21
natural materials	45	45/45	40	37	1.25	21/21
man-made materials	109	109/109	38	18	3.46	–

Dictionary capacity. A dictionary larger than the class count improves every world, and the validation choice of K is also the best test candidate in seven of the eight (Table S5). K is selected from the class count, 16, 24 and 32 by the dominant-endmember mIoU on the validation views, with the regularization and seed 42; the class map is fitted on the training views. At K equal to the class count, the regularization spends several endmembers on the appearance modes of frequent classes, so some classes receive none; a larger K relaxes this constraint. In Study and Karst, the regularization favours the appearance separation of frequent classes over rare ones (ink, soil). 3DGS with VCA/NNLS at $K = 24$ remains below Spectral World Renderer in every world, nearly tying in Study.

Table S4. **Training settings.** All methods train per scene on the same noisy 480×270 views, divided by L_{ref} and clamped to $[0, 1]$, and render at 960×540 from one stereo point cloud.

Spectral World Renderer	
RGB anchor	30k iterations; harmonic features refined for 5k steps
Decoder	30k Adam steps, 2048 pixels of one view each
Learning rate	10^{-3}, cosine decay to 10^{-5}; EMA 0.99 evaluated
Endmembers	frozen for 3k updates, then rate 10^{-3}
Assignment	weight $\max(0.005,\ 0.05(1 - t/3000))$
Stabilizers	$\hat{\mathbf{A}}$ denominator $\geq 10^{-8}$; $\mathcal{L}_p$ if $\sum_i w_i > 10^{-6}$
Clusters	≤ 2304 spectra per view, 3×3 box filter, norm > 0.03
No regularization	K = class count (SWR-Bench), $K = 16$ (real)
3DGS + VCA/NNLS	64-channel 3DGS (30k steps), VCA [59], NNLS [45]
UnMix-NeRF	official code and schedule, 30k iterations

D. Decoder Analysis

Fidelity and identification. Table S6 extends the decoder study of the main paper (Sec. 4.2) to a second scene.

- *Fidelity comes from the feature field.* A free spectral decoder in place of unmixing gains at most 0.07 dB, whereas a single feature per Gaussian costs 0.4 to 1.6 dB.
- *Identification needs harmonic encoding and intensity.* Without either, Market keeps its PSNR but loses much of its mIoU, as the mixtures absorb shading.
- *Per-intersection decoding identifies best.* Decoding after compositing fits spectra slightly better (0.1 to 0.7 dB with the full model), but per-intersection decoding keeps the higher mIoU in all three tested worlds and alone gives each Gaussian its own abundances.

What the features encode. Projected on their principal components, the features follow both material and shading (Figure S2). Within one Gaussian, intensity varies with

Table S5. **Dictionary capacity** (dominant-endmember mIoU, seed 42). Test mIoU of the regularized decoder for each candidate K; shaded: the K selected on validation views, whose validation mIoU is given in column Val. Last two columns: test mIoU at $K = 24$ without regularization and of 3DGS with VCA/NNLS. Karst includes water.

		Regularized, test					$K = 24$, test	
World	Classes	K=classes	16	24	32	Val	w/o reg.	3DGS
Forbidden City	9	0.410	0.500	0.538	0.552	0.563	0.396	0.277
Market	6	0.340	0.611	0.784	0.754	0.783	0.505	0.323
Botanical	6	0.315	0.628	0.684	0.712	0.713	0.629	0.356
NVIDIA HQ	8	0.202	0.307	0.311	0.315	0.320	0.247	0.199
Study	5	0.402	0.516	0.521	0.517	0.517	0.588	0.514
Karst	5	0.504	0.562	0.561	0.725	0.727	0.694	0.395
City	11	0.263	0.370	0.477	0.519	0.507	0.348	0.179
Office	8	0.437	0.500	0.524	0.527	0.533	0.307	0.163

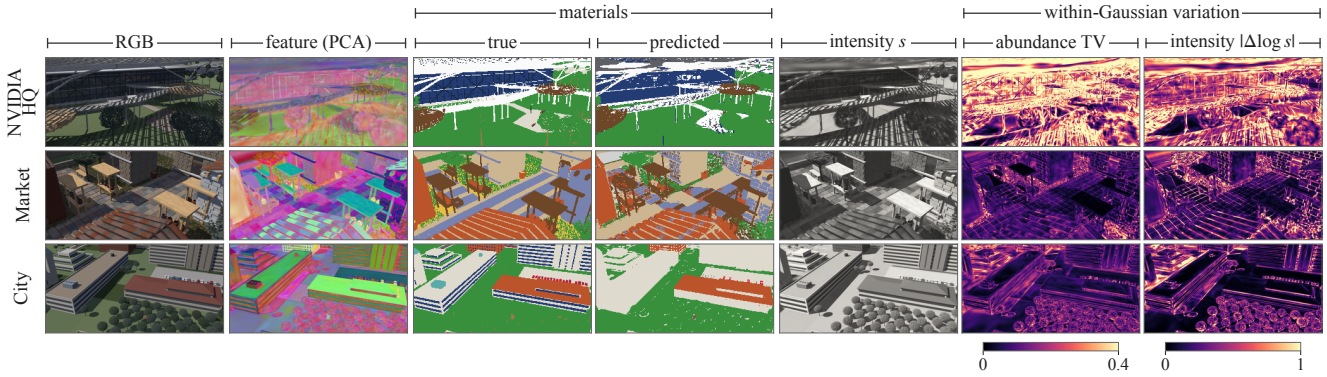


Figure S2. **Harmonic features and their decoding.** Columns show RGB, the per-hit feature (PCA), true and predicted materials, intensity, and the variation of mixture and intensity within each Gaussian against its mean feature in the same viewing direction (fixed scales).

Table S6. **Decoder controls** (no appearance regularization; means of seeds 42 and 43). Mix: unmixing decoder (✗: free decoder outputting spectra). Hit: per-intersection decoding (✗: deferred, per pixel). Vtx: four vertex features (✗: one per Gaussian). Harm.: harmonic encoding. Int.: separate intensity (✗: s=1). (h) is NHT.

						NeSpoF scene3		Market		
Exp.	Mix	Hit	Vtx	Harm.	Int.	PSNR↑	SAM↓	PSNR↑	SAM↓	mIoU↑
(a)	✓	✓	✓	✓	✓	29.62	4.22	33.72	1.87	**0.47**
(b)	✓	✓	✗	✓	✓	29.24	4.30	32.09	2.08	0.34
(c)	✓	✓	✓	✗	✓	29.61	4.21	33.65	1.89	0.30
(d)	✓	✗	✓	✓	✓	29.16	**4.17**	34.37	1.57	0.37
(e)	✓	✓	✓	✓	✗	29.54	4.37	33.70	2.14	0.22
(f)	✗	✓	✓	✓	–	**29.69**	4.28	33.74	1.72	–
(g)	✗	✓	✗	✓	–	29.26	4.34	32.11	1.90	–
(h)	✗	✗	✓	✓	–	29.21	**4.17**	**34.38**	**1.41**	–

shading, while mixture coefficients change mainly at material boundaries, so a single primitive can span an edge.

What the decoder learns. Intensity carries shading; endmembers carry material and illumination (Figure S3).

- *Intensity.* Within one specimen, the log composited intensity follows the log true radiance ($r = 0.97$ on Market, 0.87 on City), while the class mass hardly changes between sun and shadow.
- *Endmembers.* A class occupies different endmembers in sun and in shadow, which overlap by only 5 % on Market; shadow endmembers lie closer to the sky radiance than to the direct radiance. The over-complete dictionary thus represents illumination modes within a class.
- *View dependence.* On the Lambertian worlds, abundances change little with the viewing direction (total variation at most 0.04 within 45°).

Primitives as material carriers. Figure S4 draws the tetrahedron that bounds each Gaussian and colours it by the class decoded at its centre. Tetrahedra of one surface share a class, and class boundaries in the rendered map coincide with boundaries between primitives, so the abundances that a material query selects belong to individual Gaussians.

E. Real Captures

Data. HS-NeRF cubes are aligned with the released RGB images and COLMAP cameras, and only uniquely aligned views are kept; Fakeplant2 and Pinecone are excluded, as their processed RGB has a white background absent from the raw cubes. Figures S5–S7 show the first test view of every scene; errors concentrate at edges and thin parts.

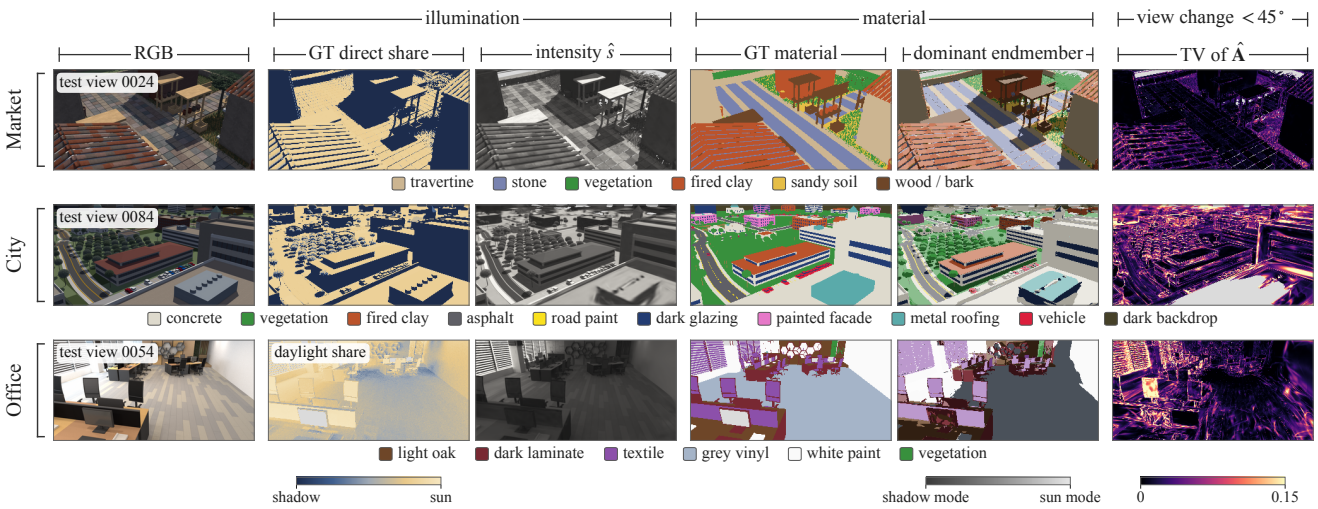


Figure S3. **Decoded intensity and endmembers.** Test views of Market (K=24), City (K=32) and Office (K=32). The composited intensity $\hat{s}$ follows the direct share of the illumination, and each material occupies distinct endmembers in sun and in shadow, so the overcomplete dictionary represents illumination modes within a class. Columns: RGB; ground-truth direct share (Office: daylight share); $\hat{s}$; ground-truth material; dominant endmember, coloured by its class and shaded by the mean direct share of its pixels; total variation (TV) of $\hat{\mathbf{A}}$ when the viewing direction changes by up to $45°$.

Table S7. **Real captures and splits.** Test views are spread along each trajectory, about one view in seven; NeSpoF also uses every fifth training view for validation. HS-NeRF sensors: Surface Optics (SO) and BaySpec (BS). Bold: totals.

Dataset / scene	Bands	Range (nm)	Resolution	Train	Test
NeSpoF	21	450–650	512 × 428		
scene1 / 2 / 3 / 4				26 / 21 / 15 / 21	4 / 4 / 3 / 4
total				**83**	**15**
HSI-SC apple	204	397–1004	512 × 512	**56**	**8**
HS-NeRF (SO)	128	371–1044	348 × 260		
Basil / Rosemary				41 / 37	6 / 6
Harsh1 / Harsh2				42 / 20	7 / 3
HS-NeRF (BS)	141	401–1087	320 × 256		
Fakeplant1				50	8
total				**190**	**30**
All real captures				**329**	**53**

F. Efficiency

Storage. Each Gaussian stores 59 values at any band count, against 78 for HyperGS and $62{+}L$ for per-band fields, so our models are 6 to 32 times smaller on the real captures, for example 4.2 MB against 127 MB on the 204-band apple.

Training and rendering. The City pipeline takes 397 s (Table S9); anchor and features do not depend on K, so tuning K repeats only the clustering and the decoder. Decoding every intersection costs more than rasterizing a stored spectrum, but our fused fp16 kernel, which skips intersections of negligible weight, renders 34.9 labelled views per second, about ten times 3DGS with VCA (Table S10).

G. Altered Leaves and Object Queries

Altered leaves. In the City variants, Spectral World Renderer separates mineral and stressed leaves from healthy foliage better than RGB clustering: the total variation between their label distributions is 0.85 and 0.83, against at most 0.40 for RGB k-means (main paper, Figure 9(c)).

Procedure. Each Gaussian is assigned its mean abundance and radiance over the training views. The type "tree" denotes crown materials and a size: Gaussians of crown material above the ground are clustered with DBSCAN [15], and large clusters form instances. A measured leaf spectrum then selects instances by comparing each crown with the query and with ordinary leaves under sunlight and skylight. All parameters were tuned on the base City.

Results. The substituted mineral is $42°$ from ordinary leaves in spectrum but only $2°$ in RGB, so RGB retrieves wrong trees and Spectral World Renderer none (Table S11; main paper, Figure 9(a,b)). The true spectra give the same answers, and the missed targets are lost earlier, mostly in crowns too sparse to form instances.

Table S11. **Object queries in City** (seeds 42 / 43). Targets: substituted (metamer) or stressed trees. Lost before the spectral query: typed as another material, or too few Gaussians to form an instance. Inst.: trees returned by the type query. Both queries retrieve every found target; wrong: ordinary trees retrieved.

		Lost				Wrong↓	
Variant	Targets	typed	sparse	Found	Inst.	SWR	RGB
Metamer	61	13 / 16	43 / 43	5 / 2	57 / 54	**0 / 0**	11 / 2
Stress	87	6 / 7	62 / 62	19 / 18	63 / 60	**1 / 1**	**1 / 1**

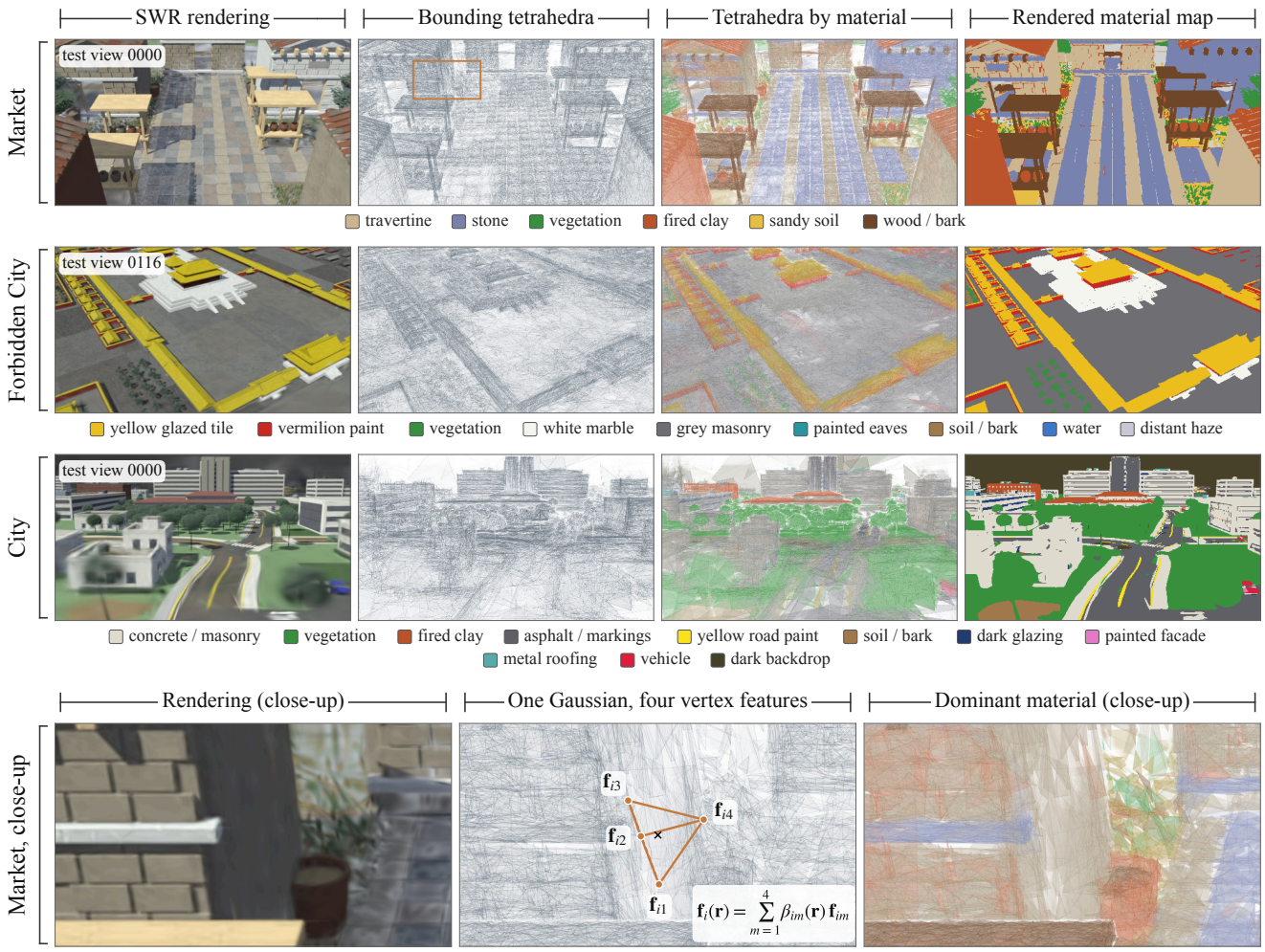


Figure S4. **Bounding tetrahedra on test views.** Each Gaussian G_i is enclosed by the regular tetrahedron of its canonical frame, scaled by $\boldsymbol{\sigma}_i$, rotated by $\mathbf{R}_i$ and translated by $\boldsymbol{\mu}_i$; the barycentric coordinates of the point of maximal response weight its four vertex features (Eq. 3 of the main paper). Columns show the rendering, the tetrahedra weighted by opacity and visibility, the tetrahedra coloured by the dominant class decoded at each Gaussian centre, and the rendered material map. The tetrahedra follow the scene structure, and their colouring reproduces the class layout of the rendered map: the median visible Gaussian places more than 98 % of its centre abundance on one class. The bottom row isolates one Gaussian of Market (amber window).

H. Query by Spectrum

Procedure. The query is a spectrum: the rendered spectrum at a clicked pixel or a library reflectance $\boldsymbol{\rho}$. Each Gaussian carries its mean spectrum over the training rays, $\bar{\mathbf{h}}_i \propto \sum_{\mathbf{r}} w_i(\mathbf{r})\,\mathbf{h}_i(\mathbf{r})$. Let $\boldsymbol{\phi}_\mathrm{d}, \boldsymbol{\phi}_\mathrm{s} \in \mathbb{R}^L$ denote the direct and sky irradiance of the world integrated over the band responses; both are known for SWR-Bench, and the cone and the whitening below use them. Under sun and sky, a surface yields radiance proportional to $\boldsymbol{\rho} \odot (\alpha \boldsymbol{\phi}_\mathrm{d} + \boldsymbol{\phi}_\mathrm{s})$, whose shape varies with the direct shading α. The comparison removes this dependence in two ways:

- a library spectrum is compared by its angle to the non-negative cone spanned by $\boldsymbol{\rho} \odot \boldsymbol{\phi}_\mathrm{d}$ and $\boldsymbol{\rho} \odot \boldsymbol{\phi}_\mathrm{s}$, which contains every sunlit, penumbral and shadowed mixture;
- a clicked spectrum is compared after illumination whitening: the RMS log-ratio of the two spectra once the constant vector and the two principal directions of $\log(\alpha \boldsymbol{\phi}_\mathrm{d} + \boldsymbol{\phi}_\mathrm{s})$, $\alpha \in [0, 1]$, are projected out. These two directions account for 99.999 % of the variation of this family.

Neither step involves training. The per-Gaussian angles are rasterized with the rendering weights; a default threshold follows from Otsu's method [61] on the histogram of log angles, and a slider adjusts it.

Evaluation. Market, City, Botanical and Karst use the models of Table 2. Clicks are three sunlit and three shadowed



Figure S5. **Qualitative comparison on HS-NeRF test views**, the first test view of each of the five scenes. Rows show false colour (630, 550 and 460 nm) and per-pixel spectral angle (left scale). Bold marks the best scores.

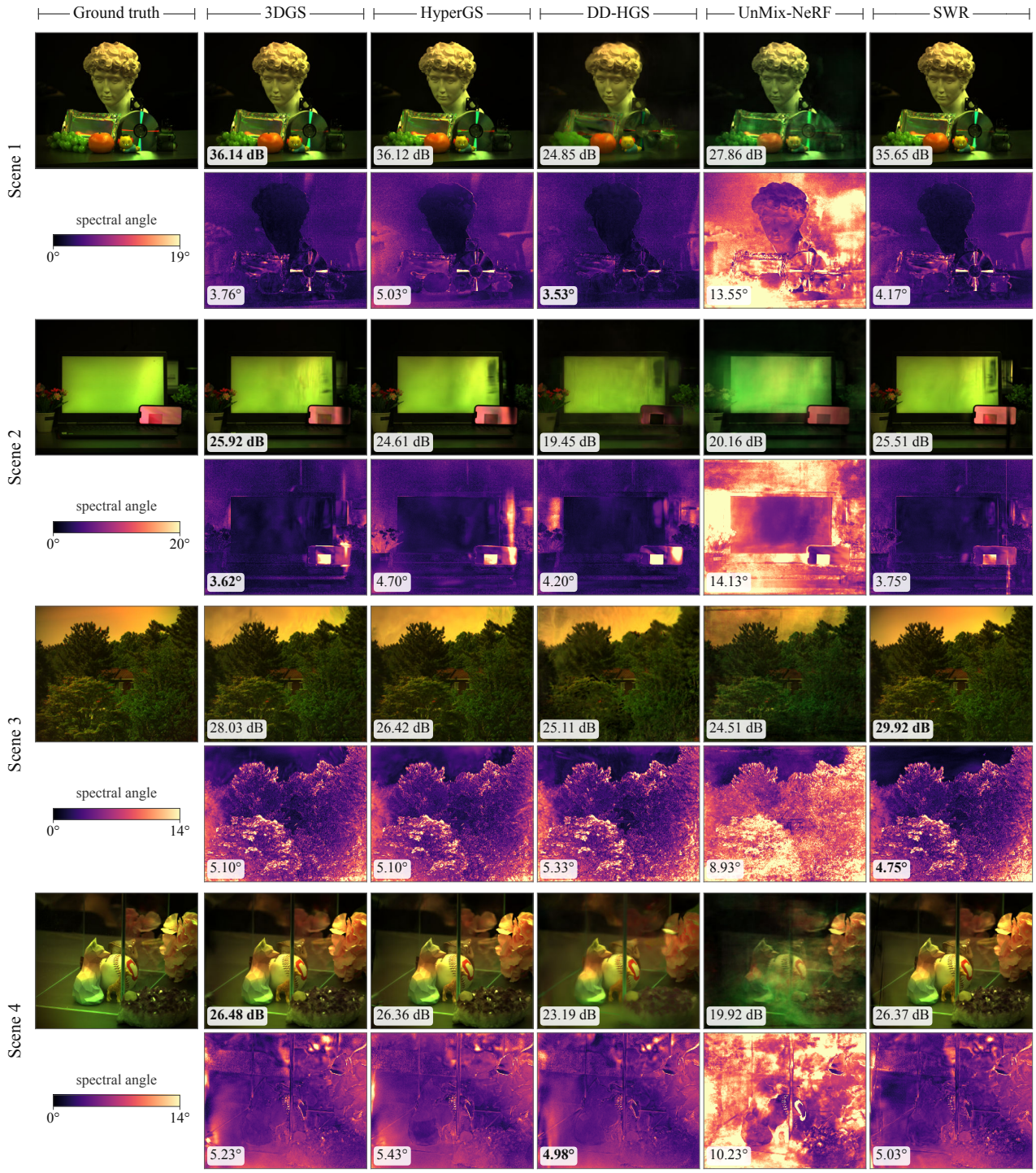


Figure S6. **Qualitative comparison on NeSpoF test views**, the first test view of each of the four scenes, as in Figure S5.

Table S8. **Spectral reconstruction on real captures, per scene** (test views; the main paper reports the means). MS/HS: measured multispectral/hyperspectral; 3DGS stores one channel per band. PSNR is higher-is-better; SAM and p95 angle are lower-is-better. Darker shading marks best, second and third; OOM is retained.

Source	Scene	Spectral World Renderer			HyperGS			DD-HGS			3DGS			UnMix-NeRF		
		PSNR	SAM	p95	PSNR	SAM	p95	PSNR	SAM	p95	PSNR	SAM	p95	PSNR	SAM	p95
MS	NeSpoF scene1	30.74	3.82	7.12	31.38	4.66	9.46	21.77	3.61	6.65	30.68	3.63	6.82	21.77	13.70	22.79
	NeSpoF scene2	28.38	3.57	8.00	26.56	4.30	9.15	21.40	3.47	7.64	27.57	3.32	7.44	21.69	14.33	22.41
	NeSpoF scene3	29.80	4.19	9.36	26.05	5.11	10.91	25.42	4.91	10.64	25.13	5.40	11.33	20.54	8.90	15.41
	NeSpoF scene4	28.64	3.85	7.66	29.44	4.27	8.24	23.09	3.44	6.82	29.07	3.68	7.41	19.36	11.56	20.40
	Mean	29.39	3.86	8.04	28.36	4.58	9.44	22.92	3.86	7.94	28.11	4.01	8.25	20.84	12.12	20.25
HS	HSI-SC apple	30.67	1.35	3.87	27.95	1.64	4.88	26.69	1.67	5.07	26.29	1.65	5.36	OOM		
HS	Basil	30.30	6.19	17.19	29.58	7.01	18.51	28.32	6.67	17.66	29.99	6.21	17.13	19.42	17.10	34.39
	Rosemary	29.21	4.30	21.48	29.05	4.60	22.35	27.70	6.96	27.18	28.57	4.43	23.31	18.67	15.82	37.50
	Harsh1	21.31	6.22	17.69	20.92	6.81	19.14	20.95	5.99	16.55	21.02	5.99	16.88	16.26	13.28	41.92
	Harsh2	24.78	6.71	17.60	23.07	6.54	16.08	22.34	5.84	14.73	23.09	6.02	15.20	15.34	21.28	51.16
	Fakeplant1	26.93	6.44	12.60	27.86	7.49	15.55	30.12	5.86	11.62	28.93	5.92	11.24	22.33	9.63	21.15
	Mean	26.51	5.97	17.31	26.10	6.49	18.33	25.89	6.26	17.55	26.32	5.71	16.75	18.40	15.42	37.22

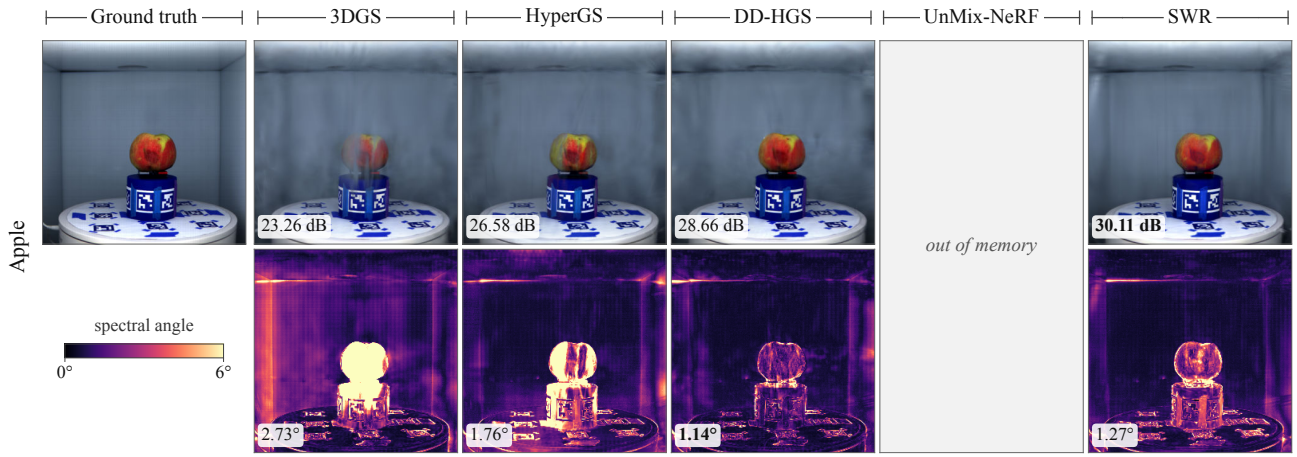


Figure S7. **Qualitative comparison on the HSI-SC apple**, first test view, as in Figure S5. UnMix-NeRF runs out of memory.

Table S9. **City training time** in seconds (one RTX 4090, seed 42, including start-up and loading; ours: mean of two runs). Ours: anchor Gaussians 73 and harmonic features 37; appearance clusters 68 and decoder 220. UnMix-NeRF trains both jointly.

Method	Representation	Unmixing	Total
UnMix-NeRF	1457		1457
3DGS + VCA/NNLS	5030	71	5102
SWR	110	288	**397**

pixels inside a single class in training views. Library queries use the assigned specimen of each class, another specimen of the class and ECOSTRESS spectra [53] that do not occur in the world. Each query is scored by AP against the class mask on all test views, and the four worlds are weighted equally. Click queries operate on rendered pixels; per-Gaussian queries differ from them by at most 0.02 AP.

Results. Illumination handling makes a measured spectrum an effective query (Figure S9).

Table S10. **Rendering speed** (one RTX 4090): test views per second at $960 \times 540 \times 64$, including spectra and labels. For 3DGS + VCA/NNLS, the second row adds the training renders that VCA needs. Ours: full precision (reference), fp16 skipping intersections of weight $< 10^{-4}$ (fast), and the same in one fused kernel; dominant labels agree with the reference on 99.97 % of pixels.

Method	Market	Botanical	Study	Karst	Mean
UnMix-NeRF	0.18	0.21	0.21	0.12	0.18
3DGS + VCA/NNLS	2.53	2.51	4.70	4.73	3.6
incl. training renders	1.49	1.47	2.18	2.16	1.8
SWR, reference	3.38	3.79	3.52	2.64	3.3
SWR, fast	8.19	8.81	8.09	6.34	7.9
SWR, fused	**36.5**	**39.3**	**36.3**	**27.4**	**34.9**

- *Shadowed clicks.* AP rises from 0.45 for RGB to 0.61, and the query retrieves 77 % of the sunlit part of the clicked material, against 10 % for RGB (Figure S8(b)).

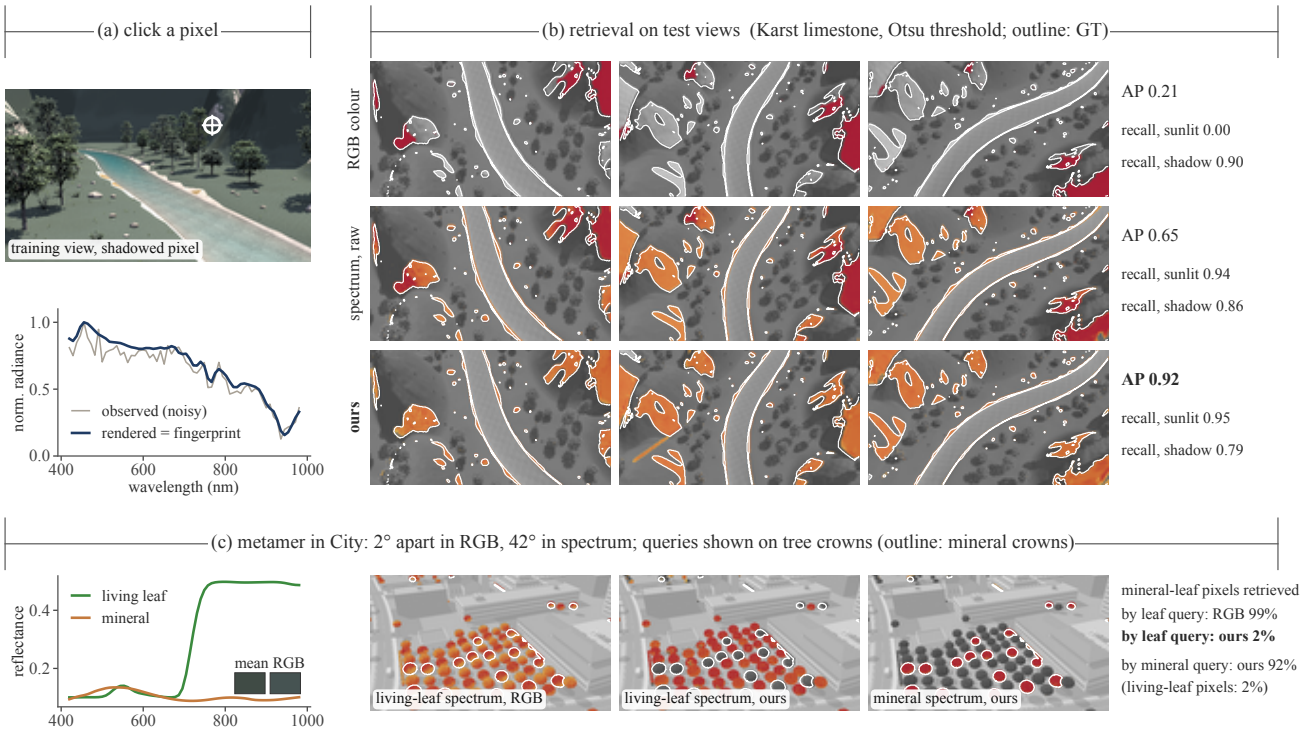


Figure S8. **Queries by a clicked or a library spectrum.** (a) A pixel on shadowed limestone is clicked in a training view of Karst; its rendered spectrum is the query. (b) Selections on three test views at the Otsu threshold for the RGB colour angle and the raw and whitened spectral angles on per-Gaussian spectra (white outline: ground truth); AP and recall cover all test views. The RGB query misses the sunlit rock, which the whitened query retrieves. (c) City metamer on tree crowns (outline: mineral crowns): the living-leaf spectrum retrieves the mineral crowns in RGB but not with Spectral World Renderer, and the mineral spectrum retrieves mainly them.

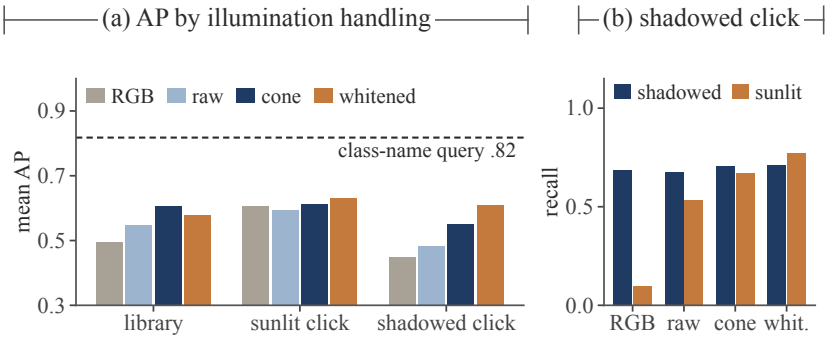


Figure S9. **Spectral queries with and without illumination handling** (mean over four worlds). (a) Mean AP of RGB queries and of spectral queries without illumination handling (raw), with the sun–sky cone and with whitening; dashed: class-name query. (b) For shadowed clicks, RGB queries retrieve the shadowed part of the class but almost none of its sunlit part; whitened spectral queries retrieve both.

- *Library spectra.* The illumination cone raises AP from 0.50 for RGB to 0.61; for sunlit clicks, spectral and RGB queries perform on par.
- *Metamers.* In the City variant, the substituted mineral is 2° from foliage in RGB but 42° in spectrum. A living-leaf spectrum separates the two leaf types (AUC 0.99 against 0.26 for RGB) and retrieves almost none of the mineral leaves, unlike RGB (Figure S8(c)).

When the material name is known, the class-name query is the more direct choice; the Otsu threshold gives the default selection, and a slider refines it.

Speed. A query, including rasterizing the selection at 960×540, takes 3 to 4 ms on an RTX 3090.

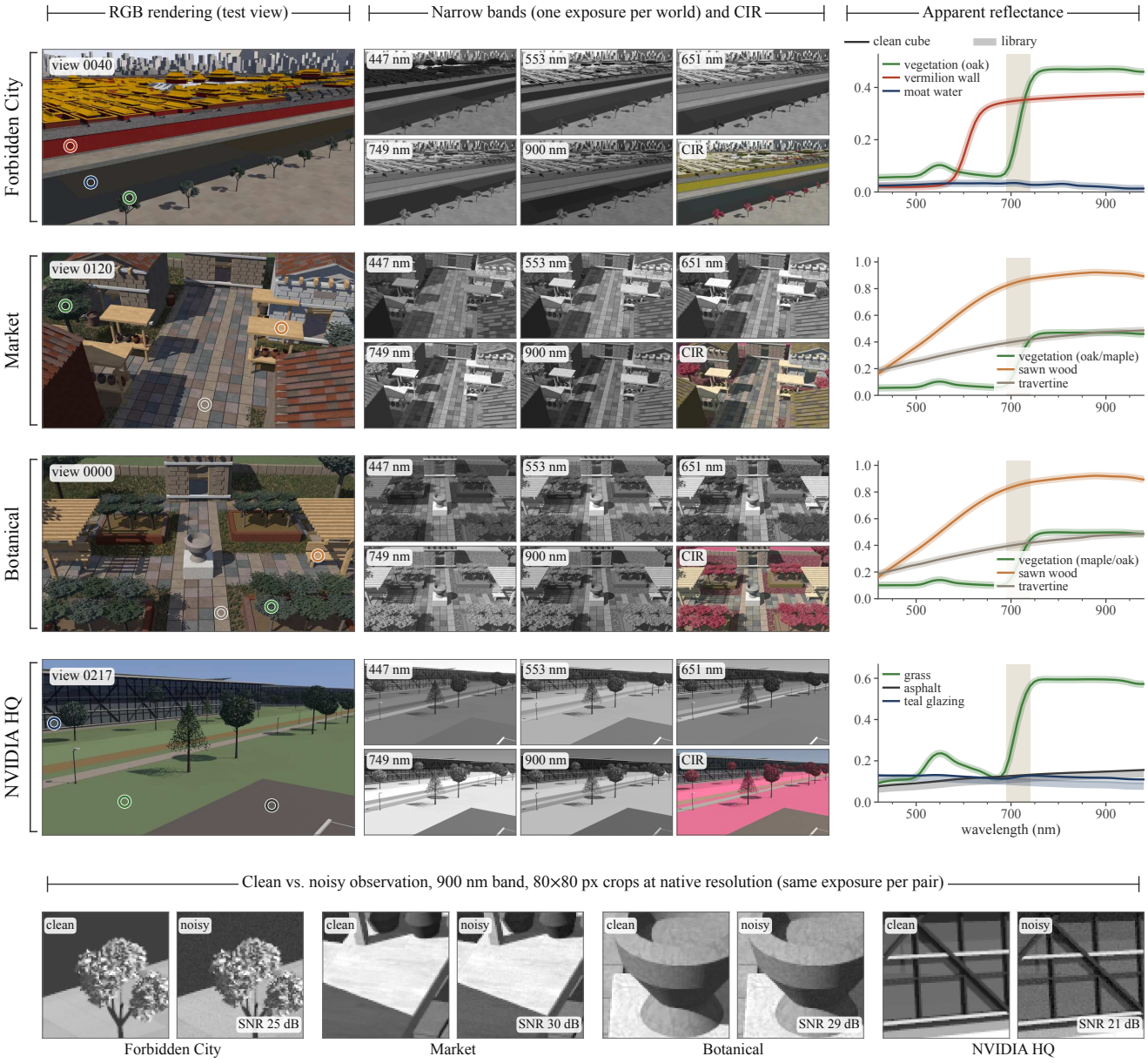


Figure S10. **Simulated hyperspectral observations: Forbidden City, Market, Botanical and NVIDIA HQ.** Narrow bands share one exposure per world, so brightness differences between bands follow the radiance; the colour-infrared composite (R, G, B = 847, 651, 553 nm) renders vegetation red. Apparent reflectance divides the clean radiance by the white-reference radiance under each pixel's illumination (median and 10–90 % range, over the library reflectance of the assigned specimen; shaded: red edge, 690–740 nm). The recovered spectra reproduce the library reflectance, including chlorophyll absorption near 670 nm and the red edge. Bottom: clean and noisy crops of the 900 nm band with the signal-to-noise ratio of the noisy crop.

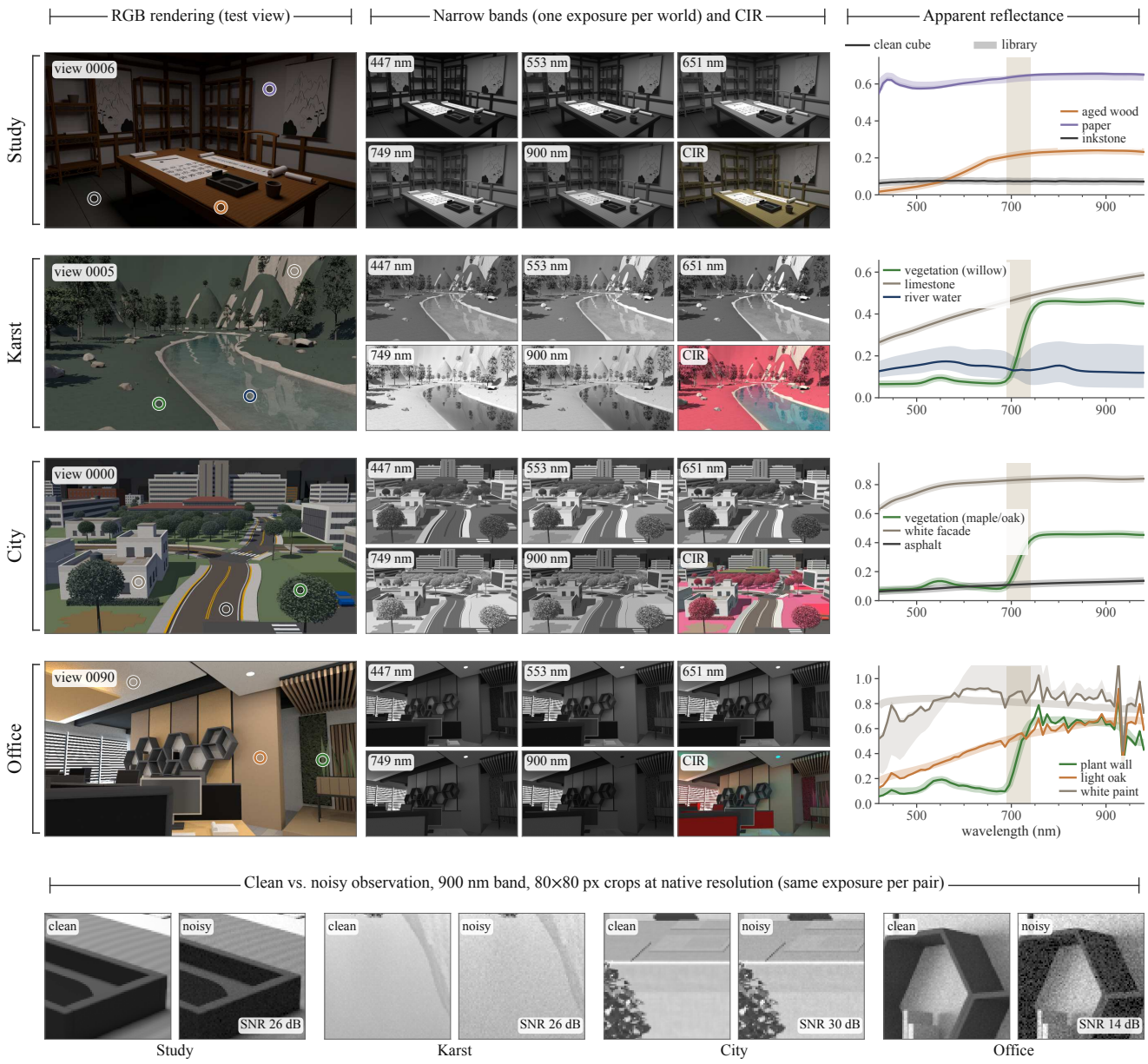


Figure S11. **Simulated hyperspectral observations: Study, Karst, City and Office.** Layout as in Figure S10. Indoor worlds are lit by white sources, so their bands differ less; the Office lights emit no near-infrared, which comes only from daylight through the blinds. Path-traced interiors deviate through interreflection and residual Monte Carlo variance.

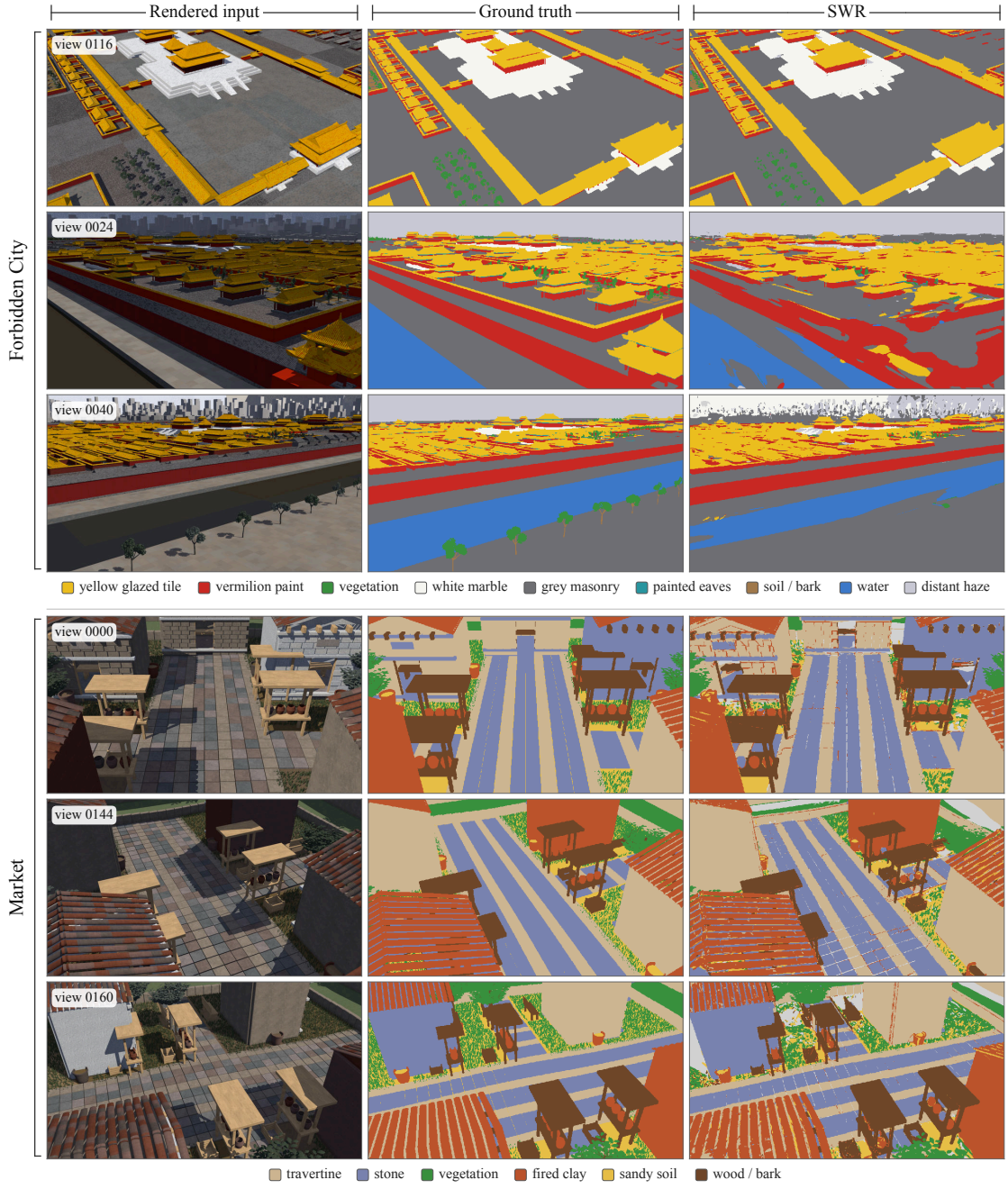


Figure S12. **Additional test views: Forbidden City and Market.** Three separated test cameras per world, shown as full frames.

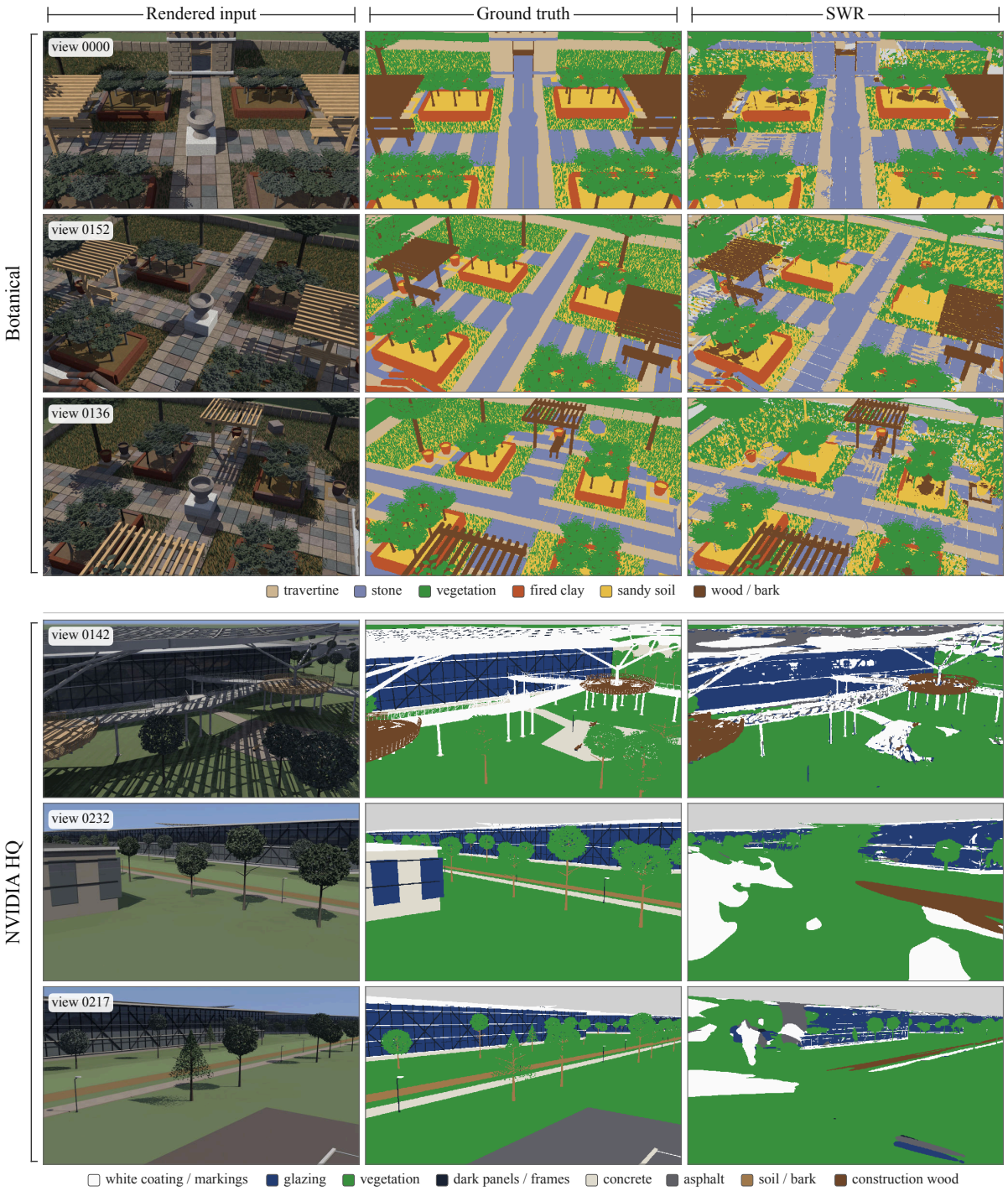


Figure S13. **Additional test views: Botanical and NVIDIA HQ.** Three separated test cameras per world, shown as full frames.

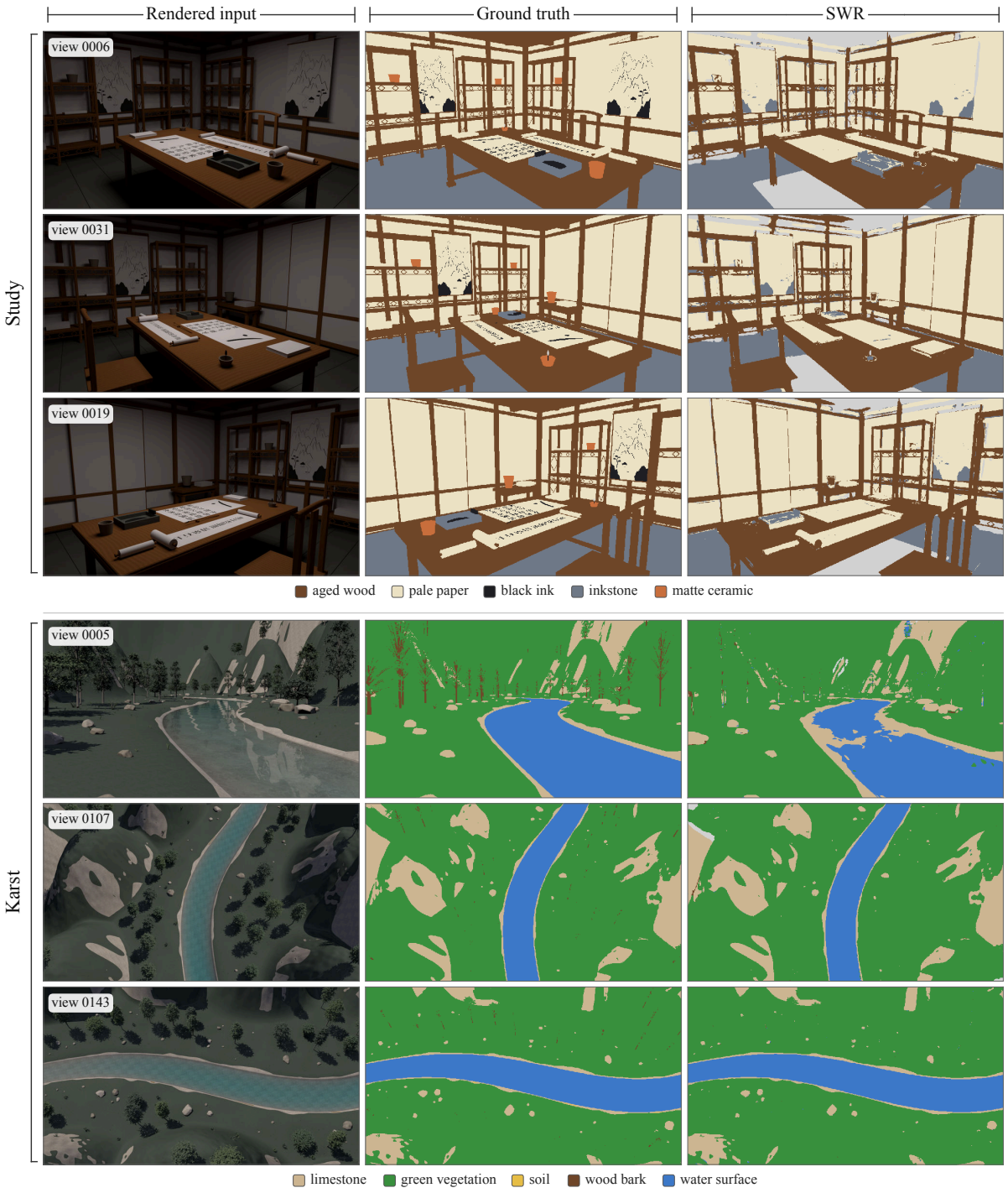


Figure S14. **Additional test views: Study and Karst.** Three separated test cameras per world, shown as full frames.

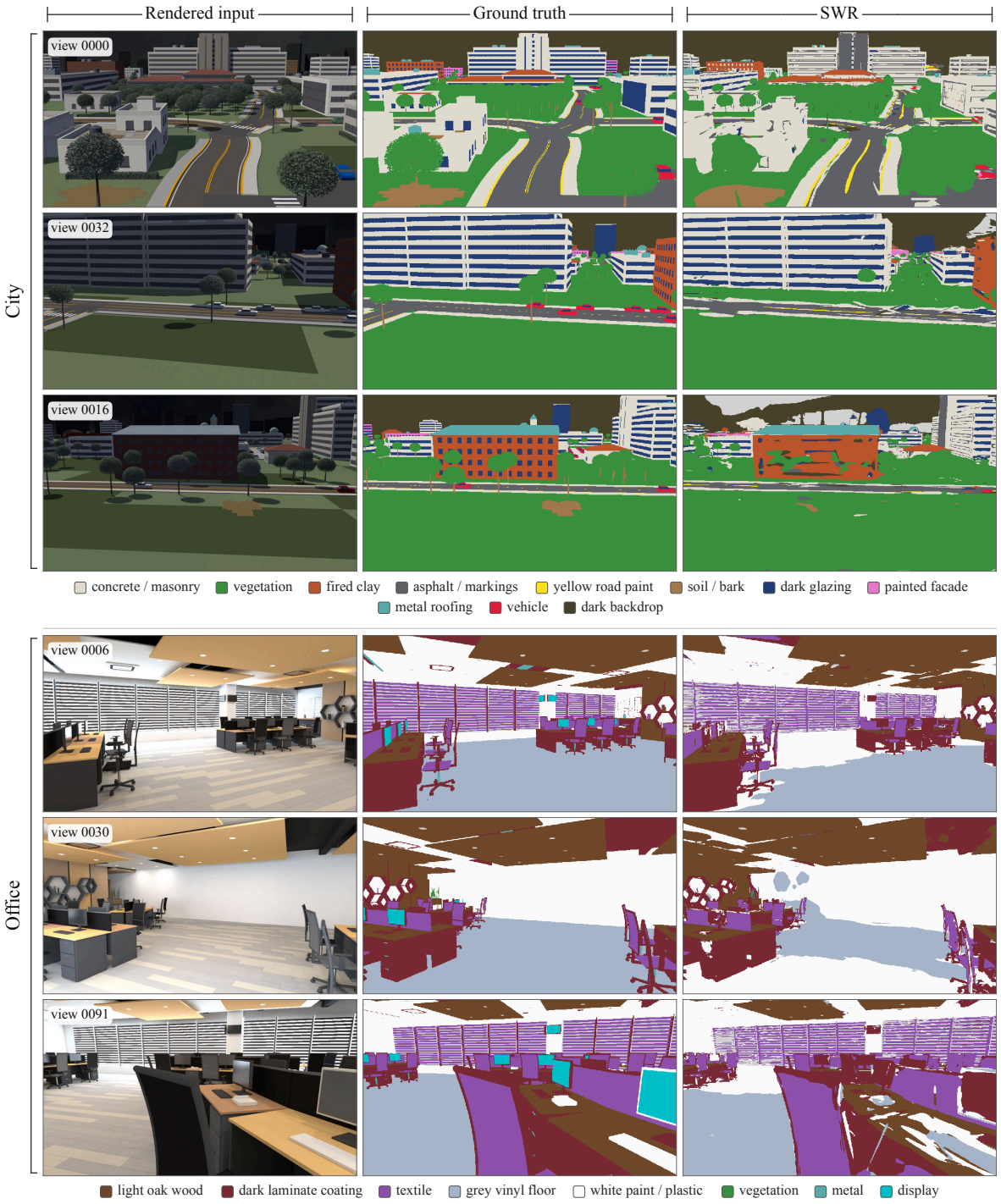


Figure S15. **Additional test views: City and Office.** Three separated test cameras per world, shown as full frames.